ROTATION INVARIANCE FROM LATTICE SYMMETRY AND REFLECTION POSITIVITY

Abstract. Consider a Gaussian-dominated random distribution with law invariant under translations. For tensor representations with planar weights of absolute value at most two, we prove that reflection positivity across the coordinate and diagonal hyperplanes implies $\mathrm{O}(d)$-covariance of the correlation functions away from coincidences. The intended application is to continuum limits of lattice theories: once translation invariance and the uniform Gaussian L^2 bound are established, rotation covariance follows from the indicated reflection positivities.

1. Introduction

Fix a number $d \geq 2$, and consider d-dimensional Euclidean spacetime $\mathbb{R}^d$. Our precise setup is described in §1.1. In particular, we have a nuclear space $\mathcal{S} = \mathcal{S}(\mathbb{R}^d; \mathcal{T})$ of test functions on $\mathbb{R}^d$ and a corresponding LCTVS $\mathcal{S}'$ of distributions. Here $\mathcal{T}$ is a finite-dimensional real orthogonal representation of $\mathrm{O}(d)$ whose complexified action in each coordinate plane has weights in $\{-2, -1, 0, 1, 2\}$.* By a "random distribution" is meant a Borel probability measure

$$\mu : \mathrm{Borel}(\mathcal{S}') \to [0, 1] \tag{1}$$

on $\mathcal{S}'$, which we assume possesses well-defined moments $S_N : \mathcal{S}^N \to \mathbb{R}$. Let $\Phi : \mathcal{S} \to \mathbb{C}$ denote the characteristic function

$$\Phi(\chi) = \int_{\mathcal{S}'} e^{iF(\chi)} \, \mathrm{d}\mu(F) \tag{2}$$

of the measure. Let $\mathcal{S}_N \subset \mathcal{S}^N$ denote the set of tuples $\chi = (\chi_1, \dots, \chi_N)$ such that the test functions $\chi_\bullet$'s have pairwise disjoint supports. The main Osterwalder–Schrader axioms [OS73] concern the behavior of $S_N(\chi)$ for $\chi \in \mathcal{S}_N$.

Let $\mathcal{N}$ denote the set of all hyperplanes $\subset \mathbb{R}^d$ of the form

$$\{x_j = 0\} \quad (0 \leq j \leq d-1), \qquad \{x_j + x_k = 0\}, \quad \{x_j - x_k = 0\} \tag{3}$$

for $0 \leq j < k \leq d-1$ in the latter two families.

Theorem A. Suppose that:

(i) There exists a constant $C > 0$ such that

$$\int_{\mathcal{S}'} e^{F(\chi)} \, \mathrm{d}\mu(F) \leq e^{C\|\chi\|_{L^2}^2} \tag{4}$$

for every real $\chi \in \mathcal{S}$.

(ii) The measure is invariant under spacetime translations. That is, for any $T : \mathbb{R}^d \to \mathbb{R}^d$ of the form $T(\mathbf{x}) = \mathbf{x} + \mathbf{a}$ for $\mathbf{a} \in \mathbb{R}^d$,

$$\Phi(T^*\chi) = \Phi(\chi). \tag{5}$$

(iii) The measure is reflection positive across each of the hyperplanes $H \in \mathcal{N}$: for any finite sequence $\chi_0, \chi_1, \dots \in \mathcal{S}$ of test functions supported on one side of $\mathbb{R}^d \backslash H$, and for any $z_\bullet \in \mathbb{C}$,

$$\sum_{a,b} \bar{z}_a z_b \Phi(\chi_a - \Theta^* \chi_b) \geq 0, \tag{6}$$

where $\Theta : \mathbb{R}^d \to \mathbb{R}^d$ denotes reflection across the hyperplane H.

*The argument allows scalars, vectors, and two-tensors, as well as higher tensors whose planar weights satisfy this restriction.

Then the measure is rotationally-invariant in the sense of Osterwalder–Schrader, meaning $S_N(R^*\chi) = S_N(\chi)$ for all $R \in \mathrm{O}(d)$ and $\chi \in \mathcal{S}_N$. ■□

Here, $R^*\chi = (R^*\chi_1, \dots, R^*\chi_N)$.

If the measure is invariant in law under translations and rotations, then it is reflection positive across any hyperplane if it is reflection positive across one. Thus, assumptions (ii) and (iii) hold for any such reflection-positive measure. The estimate (i) is the additional analytic hypothesis used below.

Item (iii) is suited to measures constructed as a continuum limit of lattice models: reflections across the hyperplanes $H \in \mathcal{N}$ are symmetries of the lattice $\mathbb{Z}^d$, and reflection positivity passes to a limit of characteristic functions.* We expect to apply this in future work.

Remark. The exponential bound eq. (4) yields the well-definedness of the S_N and their basic analytic properties. See §A. In particular, each S_N is a (possibly tensor-valued) tempered distribution on $(\mathbb{R}^d)^N$.

We conflate the tuple $\chi = (\chi_1, \dots, \chi_N)$ with the function $\chi : (\mathbb{R}^d)^N \to \mathcal{T}^{\otimes N}$ given by $\chi = \chi_1 \boxtimes \cdots \boxtimes \chi_N$.†

Remark. Reflection invariance across a hyperplane follows easily from translation invariance together with reflection positivity across the same hyperplane; see Proposition B.1. Hence the meat of the conclusion of Theorem A is rotation invariance. It therefore remains to prove covariance for $R \in \mathrm{SO}(d)$.

Remark. That the Osterwalder–Schrader axioms invoke only $\chi \in \mathcal{S}_N$, instead of arbitrary $\chi \in \mathcal{S}^N$, means they concern the behavior of the correlation functions outside of the "big diagonal"

$$\Delta = \{(\mathbf{x}_1, \dots, \mathbf{x}_N) \in (\mathbb{R}^d)^N : \mathbf{x}_j = \mathbf{x}_k \text{ for some } j \neq k\}. \tag{7}$$

We think of a point $X \in (\mathbb{R}^d)^N$ as a configuration of N labeled points $\mathbf{x} \in X$. Then $(\mathbb{R}^d)^N \backslash \Delta$ consists of the configurations of distinct points. Covariance in the sense of Osterwalder–Schrader means that the distribution

$$S_N \circ R^* - S_N \tag{8}$$

is supported on Δ for each R.‡

One can also ask about the rotation-invariance of the *measure* μ. Under the bound eq. (4), this would be equivalent to $S_N \circ R^* = S_N$ holding on all of $(\mathbb{R}^d)^N$. The theorem above allows the possibility of a non-covariant "contact term" supported on Δ. This qualification is necessary: the hypotheses permit noncovariant contact terms, as the white-noise example in §10 shows. That section also shows that the weight cutoff cannot be enlarged uniformly from two to three. The off-diagonal covariance asserted in Theorem A is the covariance condition used in the Osterwalder–Schrader reconstruction theorem, together with its other hypotheses [OS73; OS75].

Contents

1. Introduction 1
2. Reductions 4
3. Proof sketch 8
4. Consequences of reflection positivity (RP) 9
5. Extending reflection positivity 11

*Reflection positivity must still be proved for the lattice measure. The symmetry of the lattice identifies the relevant hyperplanes; it does not by itself imply the positivity inequality.

†$(\chi_1 \boxtimes \chi_2 \boxtimes \cdots)(\mathbf{x}, \mathbf{x}', \cdots) = \chi_1(\mathbf{x}) \otimes \chi_2(\mathbf{x}') \otimes \cdots$ for $\mathbf{x}, \mathbf{x}', \cdots \in \mathbb{R}^d$.

‡Near a configuration of distinct points, choose pairwise disjoint neighborhoods of its entries. Products of compactly supported tests in these neighborhoods determine a distribution there by the Schwartz kernel theorem. A partition of unity then gives the equivalence between vanishing on such products and support on Δ.

6. The analytic continuation 14
7. The growth estimate on narrow product tests 17
8. Eliminating Fourier coefficients 21
9. Conclusion 23
10. Sharpness of the conclusion and the weight restriction 23
11. A scalar lattice application 24
Appendix A. Well-definedness of S_N 25
Appendix B. RP + translation covariance ⇒ reflection covariance 26
AI statement 27
References 27

1.1. **Setup.** Abbreviate

$$t = x_0, \quad x = x_1, \quad y = (x_2, \dots, x_{d-1}) \in \mathbb{R}^{d-2}, \quad \mathbf{x} = (t, x, y). \tag{9}$$

The space $\mathcal{S} = \mathcal{S}(\mathbb{R}^d; \mathcal{T})$ is the Fréchet space of real Schwartz tests, and $\mathcal{S}'$ is its continuous dual with the weak-$*$ topology. The Euclidean inner product on $\mathcal{T}$ fixes the L^2 norm. We complexify tests and moments when taking Fourier coefficients. For an isometry $T(\mathbf{x}) = g\mathbf{x} + \mathbf{a}$, our pullback convention is

$$(T^*\chi)(\mathbf{x}) = \rho(g)\chi(T\mathbf{x}), \qquad (T_1T_2)^* = T_2^*T_1^*. \tag{10}$$

Here $\rho(g)$ is the pullback action on the tensor indices; thus $\rho(g_1g_2) = \rho(g_2)\rho(g_1)$. It is an isometry for real g. For rotations of a coordinate plane the weight assumption means

$$\|\rho(R(\theta + i\beta))v\| \le e^{2|\beta|}\|v\|. \tag{11}$$

For two-forms the exponent can be replaced by $|\beta|$.

Write $\tau_a\chi(\mathbf{x}) = \chi(\mathbf{x} - a)$, so that τ_a moves the support by a and is the pullback of translation by $-a$. On cylinder observables put

$$U(a)F(\chi) = F(\tau_a\chi). \tag{12}$$

All reflection-positivity statements concern real tests supported on one chosen side of the hyperplane. Positivity on the other side follows from reflection invariance. By Proposition B.1, every reflection in the hypotheses is a symmetry of the law. Coordinate reflections and the reflections $x_j = x_k$ generate the signed coordinate permutations. In particular, the quarter turns used below are symmetries; this is a consequence of the hypotheses, not a separate assumption.

We use

$$S_N(\chi_1, \dots, \chi_N) = \int F(\chi_1) \cdots F(\chi_N)\, \mathrm{d}\mu(F), \qquad S_0 = 1. \tag{13}$$

The Gaussian bound gives, with a constant K depending only on C,

$$|S_N(\chi_1, \dots, \chi_N)| \le (K\sqrt{N})^N \prod_{j=1}^{N} \|\chi_j\|_2. \tag{14}$$

It also makes $S_1 = 0$. These facts and the distributional interpretation are proved in §A. All subsequent products of fields lie in every finite $L^p(\mu)$.

1.2. **Literature review.** The proof develops Streater's spectrum-to-covariance method [Str72], with the finite-covariance theorem of Bros, Epstein, and Glaser [BEG67] as an analytic antecedent. Spectral analyticity first restricts the possible transformation weights; field bounds and positivity then exclude the unwanted weights. Streater proves Lorentz invariance for $P(\phi)_2$ from the forward-cone spectrum condition, local commutativity, and cutoff-uniform field and canonical-momentum bounds. The parity-based elimination of extreme degrees is also part of his method, as Osipov

explicitly records [Osi77].* Mardin and Streater [MS76, pp. 82–83] give another explicit instance of excluding transformation weights by comparing their growth with field bounds, for already covariant Fermi fields.

Coordinate and diagonal reflection positivity were already available simultaneously in Sokal's lattice construction [Sok82, pp. 328–330]. His coordinate two-point spectral representation is (2.37), and Remark 2 on p. 330 refers to a diagonal analogue. He proposes combining diagonal reflection positivity with Streater's method in §5(2), pp. 374–375, but leaves the required relativistic two-point energy bound conjectural. The derivative estimate (A.31), p. 379, is a different conclusion; its proof is referred to an unpublished manuscript. Neither that reference nor the displayed two-point representation supplies the operator inequality on arbitrary polynomial vectors required here. We prove that inequality in §4, and derive from it the translation domains used in the geometric reductions. Section 11 gives the resulting rotation implication for Sokal's normalized scalar limits.

Several analytic ingredients are established methods. The reflected Hilbert space and positive-time semigroup are the Osterwalder–Schrader construction [OS73; OS75]; the continuation of positive kernels is due to Glaser [Gla74, Theorem A]. Generating-functional estimates and repeated reflections have precedents in Fröhlich [Frö74, Lemma 3.2] and Seiler–Simon [SS76, Theorems 4.3, 7.1, and 8.2]. We give the proofs in the form needed for the present uniform L^2 bound and the partially smeared planar insertion estimate. A different model-specific use of two reflection-positive quantizations to obtain a spectrum condition appears in Jaffe, Jäkel, and Martinez [JJM13, Proposition VI.1]. We do not import a continuation domain whose derivation assumes the continuous covariance being proved.

The contribution is the resulting intrinsic Euclidean criterion for an arbitrary Borel law in the stated representation class, in every dimension $d \geq 2$. It prescribes no field equation or interaction: the operator spectrum condition, analytic estimates, and covariance conclusion follow from the three stated assumptions. This is an adaptation of the preceding methods, rather than a claim of novelty for finite-degree reduction, parity, or growth exclusion as principles. The uniform Gaussian bound is strong: even for a centered Gaussian law, it requires the covariance form to be bounded on L^2. We claim neither minimality of the assumptions nor a uniform weakening of all earlier hypotheses. The examples in §10 identify limitations of the criterion.

2. Reductions

For each pair $j < k$, consider the subgroup $G \subset \mathrm{SO}(d)$ consisting of orthogonal transformations R that mix the two coordinates x_j, x_k among themselves and leave the other coordinates fixed. Because $\mathrm{SO}(d)$ is generated by the union of such subgroups, in order to prove full rotation covariance it suffices to prove covariance for transformations lying in each individual G, that is

$$S_N(\chi) = S_N(R^*\chi) \tag{15}$$

for all $N \in \mathbb{N}$, $R \in G$, and $\chi \in \mathcal{S}_N$. The hypotheses of the theorem are invariant under a permutation of the coordinates, so it suffices to consider the group

$$G = \left\{ R(\theta) = \begin{bmatrix} r(\theta) & 0 \\ 0 & I_{d-2} \end{bmatrix} : \theta \in \mathbb{R} \right\} \cong \mathrm{SO}(2), \qquad r(\theta) = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \tag{16}$$

of orthogonal transformations of the tx-plane.

For each N-tuple $\chi = (\chi_1, \dots, \chi_N) \in \mathcal{S}_N$ and $\theta \in \mathbb{R}$, define

$$D_N[\chi](\theta) = S_N(R(\theta)^*\chi) - S_N(\chi) \tag{17}$$

*Our attribution of the scalar parity step follows Osipov's account of Streater's proof. We make no separate priority claim for the precise cross-order inequality or the four-mirror comparison below; an equation-by-equation comparison with Streater's proof has not been established here.

to measure the failure of S_N to be rotation-invariant. Because S_N is a distribution, $D[\chi] \in C^\infty(\mathbb{S}^1_\theta)$.* Note that $D_N[\chi](0) = 0$. We want to show that

$$D_N[\chi](\theta) = 0, \tag{18}$$

for all θ.

Let $\Delta_2 \subset (\mathbb{R}^d)^N$ be the set of configurations X of points $\mathbf{x} = (t, x, y)$ such that the $(t, x) \in \mathbb{R}^2$ are not all distinct. In other words, if we let $\Pi : \mathbb{R}^d \to \mathbb{R}^2$ denote projection onto the first two coordinates, then $X \in (\mathbb{R}^d)^N$ lies outside Δ_2 if and only if all points $\mathbf{x} \in X$ have distinct projections $\Pi\mathbf{x}$. We prove:

Theorem B. Let $\mu : \mathrm{Borel}(\mathcal{S}') \to [0, 1]$ denote a Borel probability measure on $\mathcal{S}'$. Denote its characteristic function by Φ. Suppose that μ satisfies the following three properties:

(i) There exists a constant $C > 0$ such that

$$\int_{\mathcal{S}'} e^{F(\chi)} \, \mathrm{d}\mu(F) \leq e^{C\|\chi\|^2_{L^2}} \tag{19}$$

for every real $\chi \in \mathcal{S}$.

(ii) The measure is invariant under spacetime translations. That is, for any $T : \mathbb{R}^d \to \mathbb{R}^d$ of the form $T(\mathbf{x}) = \mathbf{x} + \mathbf{a}$ for $\mathbf{a} \in \mathbb{R}^d$,

$$\Phi(T^*\chi) = \Phi(\chi). \tag{20}$$

(iii) The measure is reflection positive across the four hyperplanes

$$\{t = 0\}, \quad \{x = 0\}, \quad \{t + x = 0\}, \quad \{t - x = 0\}. \tag{21}$$

Then $D_N(\theta)$ vanishes on $(\mathbb{R}^d)^N \backslash \Delta_2$ for each $\theta \in \mathbb{R}$. ■□

The conclusion of this theorem is not full Osterwalder–Schrader covariance, because it leaves open the possibility that D_N be nonzero on $\Delta_2 \supseteq \Delta$, rather than only Δ. In order to deduce full Osterwalder–Schrader covariance (under the hypotheses of Theorem A), an argument is required. The argument exploits some real-analyticity of $D_N[X](\theta)$'s dependence on X to deduce vanishing for $X \in \Delta_2 \backslash \Delta$ from vanishing for $X \notin \Delta_2$. We will show that $D_N[X](\theta)$ depends real-analytically on X in certain preferred directions. (Making this precise requires smearing against X.) Consequently, if $D_N[X](\theta)$ vanishes in an open neighborhood of some configuration $X \in (\mathbb{R}^d)^N \backslash \Delta$, then it also vanishes in some open neighborhood of configurations X' that can be reached from X by following preferred lines in $(\mathbb{R}^d)^N \backslash \Delta$.

To make this more precise, we define for each $\theta \in \mathbb{R}$ an equivalence relation $\sim_\theta$ on $(\mathbb{R}^d)^N \backslash \Delta$ as follows. Suppose $X_0, X_1 \in (\mathbb{R}^d)^N \backslash \Delta$ are two configurations for which there exists a one-parameter group of translations $\mathbb{R} \ni s \mapsto T(s)$ of $(\mathbb{R}^d)^N$ with the following properties:

- $T(1)X_0 = X_1$
- for $t \in [0, 1]$, $T(t)X_0$ stays inside of $(\mathbb{R}^d)^N \backslash \Delta$,
- for some open neighborhood $O \ni X_0$, if $\chi \in \mathcal{S}_N$ satisfies $\operatorname{supp} \chi \subset O$, then the function

$$s \mapsto D_N[T(-s)^*\chi](\theta) \tag{22}$$

is real-analytic on some neighborhood of $[0, 1]$.

Then we write $X_0 \sim_\theta X_1$. This defines a reflexive symmetric relation, which we extend to an equivalence relation by taking the transitive closure. Then, if $X_0 \sim_\theta X_1$ and $D_N[X](\theta)$ vanishes near $X = X_0$, it also vanishes near $X = X_1$.

Given a configuration $X_0 \in (\mathbb{R}^d)^N \backslash \Delta$ let

$$H \subset \mathbb{R}^d \tag{23}$$

*The notation $D_N[X](\theta)$ below is distributional shorthand: to say that it vanishes near X means that the distribution $D_N(\theta)$ annihilates every test supported in some neighborhood of X. No pointwise evaluation at X is assumed.

denote an affine hyperplane parallel to the coordinate axes and containing none of the points $\mathbf{x} \in X_0$. Then, H divides the points in X_0 into two groups. Let $\mathbf{n}$ denote a unit vector normal to H. Use that vector to designate one side of H as the forward direction. The following lemma states that $S_N(X)$ is real analytic at $X = X_0$ in any direction moving only the points in the group on the forward side of the hyperplane. More precisely, letting H_0 be the hyperplane parallel to H and passing through the origin:

Lemma 2.1. Fix X_0 as above. For $\delta\mathbf{x} \in \mathbb{R}^d$, let $\tau_{\delta\mathbf{x}} : \mathbb{R}^d \to \mathbb{R}^d$ denote the translation

$$\tau_{\delta\mathbf{x}} : \mathbf{x} \mapsto \mathbf{x} + \delta\mathbf{x} \tag{24}$$

and let $T_\bullet = T_\bullet[H] : (\mathbb{R}^d)^N \to (\mathbb{R}^d)^N$ denote the translation that acts via $\tau_\bullet$ on the jth $\mathbb{R}^d$ factor of $(\mathbb{R}^d)^N$ if the jth point $\in X_0$ lies on the designated side of H (and leaves the other factors fixed.) Then, under the hypotheses of Theorem A, there exists an open neighborhood $O \ni X_0$ such that if $\chi \in \mathcal{S}_N$ is supported in O, then the function

$$(-\varepsilon, \infty) \times H_0 \ni (s, \mathbf{v}) \mapsto S_N(T^*_{-s\mathbf{n}-\mathbf{v}}\chi) \tag{25}$$

is real analytic on $(-\varepsilon, \infty) \times H_0$ for some $\varepsilon > 0$. Under the hypotheses of only Theorem B, the same conclusion holds if the hyperplane is one of constant t or x and $\mathbf{v}$ stays in the tx-plane. ■

The lemma is proved in §6.1. Given the lemma above:

Proof of Theorem A (from Theorem B). Assume the hypotheses of Theorem A. Then Theorem B applies, so $D_N[X](\theta)$ vanishes for $X \in (\mathbb{R}^d)^N \backslash \Delta_2$. Consider a configuration $X \in \Delta_2 \backslash \Delta$. We prove that there exists some $X_0 \in (\mathbb{R}^d)^N \backslash \Delta_2$ such that

$$X \sim_\theta X_0 \tag{26}$$

for all $\theta \in \mathbb{R}$. In light of the discussion above, this will complete the proof: the vanishing of D_N near X_0 implies the vanishing near X.

For each $\# = 0, \dots, 1 + N(N-1)/2$, let $\Delta_2^\#$ denote the set of configurations $X \in (\mathbb{R}^d)^N$ such that either $X \in \Delta$ or there exists a set of at least $\#$ pairs of distinct $\mathbf{x}, \mathbf{x}' \in X$ such that $\Pi\mathbf{x} = \Pi\mathbf{x}'$. Then

$$(\mathbb{R}^d)^N = \Delta_2^0 \supseteq \Delta_2 = \Delta_2^1 \supseteq \cdots \supseteq \Delta_2^{1+N(N-1)/2} = \Delta. \tag{27}$$

Given $X \in (\mathbb{R}^d)^N \backslash \Delta$, let $\#$ be the smallest nonnegative integer such that $X \in \Delta_2^\# \backslash \Delta_2^{\#+1}$. If $\# = 0$, then $X \in (\mathbb{R}^d)^N \backslash \Delta_2$ already. If instead $\# \geq 1$, let $\mathbf{x}, \mathbf{x}' \in X$ denote two points in the configuration whose first two coordinates agree:

$$\Pi\mathbf{x} = \Pi\mathbf{x}'. \tag{28}$$

Because $X \notin \Delta$, the points $\mathbf{x}, \mathbf{x}'$ are distinct, so they must be distinguished by one of the remaining cartesian coordinates $y_1, \dots, y_{d-2}$. Now pick a hyperplane $H = \{y_\bullet = a\}$ not containing any of the points in X and separating $\mathbf{x}, \mathbf{x}'$.

Above, we defined $T_\bullet$. Redefine this with the coordinate $y_\bullet$ in place of t, using the hyperplane H to define the two groups of points. Thus, for each $\delta t, \delta x \in \mathbb{R}$, we have a translation $T_{\delta t, \delta x}$ (that does nothing to the y coordinates),

$$T_{\delta t, \delta x} : (t, x, y) \mapsto (t + \delta t, x + \delta x, y). \tag{29}$$

Now note that if δt is in some small punctured interval $(-\varepsilon, \varepsilon) \backslash \{0\}$, then

$$T_{\delta t, 0} X \in (\mathbb{R}^d)^N \backslash \Delta_2^\# \tag{30}$$

because it removes the coincidence between $\mathbf{x}, \mathbf{x}'$ by changing the time coordinate of one but not the other (and as long as ε is sufficiently small, no more coincidences are created). We will show that

$$X \sim_\theta T_{\delta t, 0} X. \tag{31}$$

If we have already proved the claim for smaller values of #, then $T_{\delta t,0}X \sim_\theta X_0$ for some $X_0 \in (\mathbb{R}^d)^N \backslash \Delta_2$, hence $X \sim_\theta X_0$.

In terms of S_N,

$$\begin{aligned} D_N[T^*_{-\delta t,0}\chi](\theta) &= S_N[R(\theta)^* T^*_{-\delta t,0}\chi] - S_N[T^*_{-\delta t,0}\chi] \\ &= S_N[T^*_{-(\cos\theta)\delta t,-(\sin\theta)\delta t} R(\theta)^*\chi] - S_N[T^*_{-\delta t,0}\chi]. \end{aligned} \tag{32}$$

Applying Lemma 2.1 with $y_\bullet$ in place of t, we conclude that the right-hand side above depends real analytically on δt near $\delta t = 0$, as long as χ is supported sufficiently close to the configuration X. Hence $X \sim_\theta T_{\delta t,0}X$. □

The result can be reduced further.

Say that a configuration $X \in (\mathbb{R}^d)^N \backslash \Delta_2$ is *narrow* if the points in X can be listed in some order $\mathbf{x}_1, \dots, \mathbf{x}_N$ such that, for some unit vector $\mathbf{n} \in \mathbb{R}^2$, the differences $\Pi(\mathbf{x}_1 - \mathbf{x}_2), \cdots, \Pi(\mathbf{x}_{N-1} - \mathbf{x}_N) \in \mathbb{R}^2$ all make angle $< \pi/4$ with $\mathbf{n}$. That is,

$$\frac{\mathbf{n} \cdot \Pi(\mathbf{x}_n - \mathbf{x}_{n+1})}{|\Pi(\mathbf{x}_n - \mathbf{x}_{n+1})|} > \frac{1}{\sqrt{2}} \tag{33}$$

for $n = 1, \dots, N-1$.

Proposition 2.2. Under the hypotheses of Theorem B, for any $X \in (\mathbb{R}^d)^N \backslash \Delta_2$, there exists a $\delta > 0$ and narrow X_0 such that $X \sim_\theta X_0$ whenever $-\delta < \theta < \delta$. ■

Proof. First, we show that every $X \in (\mathbb{R}^d)^N \backslash \Delta_2$ is equivalent to a configuration in which all points have a distinct time coordinate. Indeed, proceed via induction on the number of pairs of points sharing a time coordinate. If two times agree, their x coordinates differ. A line $x = a$ separating them and avoiding the configuration divides the points into two groups. Move one group by a sufficiently small amount in the time direction. At least one time coincidence disappears and no new one is created. The x gap remains positive. By Lemma 2.1, the moment is analytic along this move. For $|\theta|$ small, the rotated configuration and the rotated move still preserve an x gap, so the other term in D_N is analytic as well. Thus this move is allowed in $\sim_\theta$.

Now suppose that $X \in (\mathbb{R}^d)^N \backslash \Delta_2$ contains points with distinct time coordinates. Then by a series of $T_{\delta t,0}$ for different hyperplanes, we can stretch X in the vertical direction until the result is a narrow configuration: cut between successive times and move the later group forward until every successive time gap exceeds the absolute horizontal difference. Every move preserves a positive time gap. Each path is compact, so the same is true after rotation through sufficiently small $|\theta|$. Apply Lemma 2.1 to both terms of D_N along each path, and take the smallest of the finitely many allowed angle bounds. This gives the claimed δ. The same choices work for configurations in a sufficiently small neighborhood of X. □

We now state:

Theorem C. Under the hypotheses of Theorem B, $D_N[X](\theta)$ vanishes on the set of narrow configurations X. ■□

Given the theorem above:

Proof of Theorem B from Theorem C. Given the hypotheses of the theorem, Theorem C says that $D_N[X](\theta)$ vanishes whenever X is narrow. By Proposition 2.2, we conclude that given any $X \in (\mathbb{R}^d)^N \backslash \Delta_2$, there exists an open neighborhood $O \ni X$ and an interval's worth of θ such that $D_N[\chi](\theta) = 0$ whenever $\chi \in C^\infty_c(O)$.

Lemma 2.3 completes the proof. □

Given $X \in (\mathbb{R}^d)^N$, let $\mathcal{O}(X) \subset (\mathbb{R}^d)^N$ denote the set of configurations arising by applying to X a common rotation $R(\theta)$ to each point in the configuration. That is, $\mathcal{O}(X)$ is X's orbit under the group of rotations of the tx-plane.

Lemma 2.3. In order to show that $D_N(\theta)$ vanishes near some configuration $X \in (\mathbb{R}^d)^N \backslash \Delta$ for each $\theta \in \mathbb{R}$, it suffices to show that for each $X' \in \mathcal{O}(X)$ there exists an open neighborhood $O \ni X'$ with the following property: there exists a $\delta = \delta(O) > 0$ such that $D_N[\chi](\theta) = 0$ whenever $-\delta < \theta < \delta$ and $\chi \in \mathcal{S}_N$ is supported in O. ■

That is, it suffices to test small angles.

Proof. Suppose that the desired neighborhoods exist. Because $\mathcal{O}(X)$ is compact, a finite collection of the given neighborhoods covers it. Shrink $O_0 \ni X$ and take the minimum of their angle bounds so that every rotated copy of O_0 lies in one of them.* Then the choice

$$O(X') = R(-\phi)O_0 \tag{34}$$

works for $X' = R(-\phi)X$, with a common angle bound $\delta > 0$. Now fix $\chi \in \mathcal{S}_N$ supported in O_0, and consider the set

$$\{\theta \in \mathbb{R} : D_N[\chi](\theta) = 0\}. \tag{35}$$

This is closed (because $\theta \mapsto D_N[\chi](\theta)$ is continuous) and contains $\theta = 0$. To see that it is open, hence all of the real line, suppose that θ_0 lies in this set and θ satisfies $|\theta - \theta_0| < \delta$. By definition,

$$D_N[\chi](\theta) = D_N[R(\theta_0)^* \chi](\theta - \theta_0) + D_N[\chi](\theta_0). \tag{36}$$

Both terms on the right-hand side are zero. □

The rest of this paper is devoted to the proofs of Lemma 2.1 and Theorem C.

3. Proof sketch

Expanding in Fourier series, write

$$D_N[\chi](\theta) = \sum_{\ell \in \mathbb{Z}} c_\ell^{(N)}[\chi] e^{i\ell\theta}, \qquad c_\ell^{(N)}[\chi] = \frac{1}{2\pi} \int_0^{2\pi} D_N[\chi](\theta) e^{-i\ell\theta} \, \mathrm{d}\theta. \tag{37}$$

For real tests $c_{-\ell}[\chi] = \overline{c_\ell[\chi]}$; for complex tests one also conjugates χ. We suppress the dependence on N, χ when possible. Our goal is to show that $c_\ell = 0$ for all $\ell \neq 0$. Then $D_N[\chi](\theta) = c_0$ is constant, and since $D_N[\chi](0) = 0$, the constant is zero as well.

The argument is based on holomorphy in the angle. The first reflection constructs the Osterwalder–Schrader Hilbert space; the second gives $H \geq |P|$. This supplies complex translations in both t and x. Gaussian domination controls insertions between these translations. We first continue the kernel along narrow configurations, where the whole complex rotation orbit stays in the available domains, and obtain

$$|D_N[\chi](\theta + i\beta)| \leq C_\chi e^{3N|\beta|}. \tag{38}$$

The exponent consists of one inverse planar separation per insertion and at most two tensor weights per field. A contour shift therefore leaves only $|\ell| \leq 3N$. The quarter turn is a symmetry, so only $\ell \in 4\mathbb{Z}$ can occur.

We use the finite expansion only near narrow configurations, as required for Theorem C. Define M_N to be the largest nonzero absolute frequency occurring there, or zero if none occurs. Thus $M_0 = 0$ and $0 \leq M_N \leq 3N$. A common rotation puts a narrow configuration in the cone $\Delta t > |\Delta x|$. We can then place any chosen gap across the reflection plane. The support conditions in §7.1 ensure that the resulting mixed pairing and its two squared norms are all narrow. This lets the positivity argument stay within the class controlled by the growth estimate.

*Apply compactness to the parameter circle: each angle has an interval on which the corresponding rotated neighborhood remains in one fixed open set. A finite subcover gives both a common shrinkage of O_0 and a positive minimum angle bound.

Reflection positivity turns continued moments into positive forms. Cauchy–Schwarz compares a moment of order $a+b$ with squared norms of orders $2a$ and $2b$, giving

$$2M_{a+b} \le M_{2a} + M_{2b}. \tag{39}$$

The leading coefficient in an even-order squared norm is nonnegative, and for some tests it is strictly positive. Rotating the law by 45° multiplies that coefficient by $(-1)^{M_{2a}/4}$, while leaving it nonnegative in the second reflection reconstruction. Hence $M_{2a} \in 8\mathbb{Z}$. Convexity and the bound $M_N \le 3N$ then force every M_N to vanish. This proves Theorem C; the reductions above then give the main theorem.

4. Consequences of reflection positivity (RP)

4.1. **The Osterwalder–Schrader Hilbert space.** The reflected quotient and translation semigroup are the Osterwalder–Schrader construction [OS73; OS75]. We recall the details without assuming continuous rotation covariance.

Let $L^2_+(\mathcal{S}', \mu)$ denote the closed subspace of $L^2(\mathcal{S}', \mu)$ generated by observables depending on fields in the forward half-space $\{t > 0\}$. In particular, it contains all products

$$\mathrm{eval}_\chi(F) = F(\chi_1) \cdots F(\chi_N) \tag{40}$$

with the tests supported there, and $\mathrm{eval}_\varnothing = 1$. We include the constant in the algebra of such products. These polynomials are dense in L^2_+.*

Write $\Theta_\circ(t, x, y) = (-t, x, y)$ and set $J_\circ A = \overline{A(\Theta_\circ^* F)}$. Reflection invariance makes $J_\circ$ an antiunitary involution on $L^2(\mu)$. Define

$$\langle A, B\rangle_\circ = \mathbb{E}_\mu[(J_\circ A)B]. \tag{41}$$

The form is antilinear in its first argument. Reflection positivity is equivalent to its being positive semidefinite on L^2_+. Indeed, on field exponentials its matrix is the transpose of the characteristic-functional matrix in Item (iii), and the transpose of a positive Hermitian matrix is positive. The estimate

$$|\langle A, B\rangle_\circ| \le \|A\|_{L^2(\mu)} \|B\|_{L^2(\mu)} \tag{42}$$

extends positivity to the closure. Conversely, applying positivity to finite sums of exponentials gives the original condition.

Quotient by the null space and complete to obtain $\mathcal{H}_\circ$. Denote the class of A by $[A]$, and put $\Omega = [1]$. The class map is a contraction. All later Hilbert-space norms refer to this quotient, not to the ordinary $L^2(\mu)$ norm.

Let $T_{\delta t, \delta x} : (t, x, y) \mapsto (t + \delta t, x + \delta x, y)$ denote a translation fixing y. Its action on observables will mean $U(\delta t e_0 + \delta x e_1)$. For $s \ge 0$, reflection and translation give

$$\|[U(se_0)A]\|^2 = \langle [A], [U(2se_0)A]\rangle \le \|[A]\| \, \|[U(2se_0)A]\|. \tag{43}$$

Iterating and using $\|[U(re_0)A]\| \le \|A\|_{L^2(\mu)}$ proves contractivity and preservation of null vectors. The induced operators are self-adjoint and strongly continuous. Translations parallel to $t = 0$ induce commuting strongly continuous unitary groups. Therefore there are a nonnegative self-adjoint H and a self-adjoint P with joint projection-valued measure Π such that

$$V(\delta t, \delta x) = e^{-\delta t H + i \delta x P} = \int_{\mathbb{R}^2} e^{-\omega \delta t + i p \delta x} \, \mathrm{d}\Pi(\omega, p), \qquad \delta t \ge 0. \tag{44}$$

They implement the translations of classes. Strong continuity follows first on field exponentials from continuity of their tests and the field L^2 bound, and then by density. The vacuum is fixed by every

*The Gaussian bound allows the Taylor polynomials of $e^{iF(\chi)}$ to converge in L^2. Field exponentials have dense span: a function orthogonal to them has finite-dimensional conditional measures with zero Fourier transform, hence zero conditional expectations. Evaluations generate the sigma-algebra; see §A.

translation. The same construction applies with any of the assumed reflection hyperplanes as time zero.

4.2. **Spectral inequality.** Write $\Theta_\Delta(t,x,y) = (-x,-t,y)$ and J_Δ for its antilinear reflection. Now consider the Hamiltonian H and the x-momentum generator P, constructed in the previous subsection as self-adjoint operators on the Osterwalder–Schrader Hilbert space.

Proposition 4.1. $H \geq |P|$ ■

Proof. Let $A \in L^2_+$ denote a real polynomial in compactly supported field variables and $T = U(\delta t e_0 + \delta x e_1)$ a translation with $\delta t > 0$. Consider the quantity

$$\mathbb{E}_\mu[(J_\circ A)TA] = \langle [A], [TA]\rangle_{\mathcal{H}_\circ}. \tag{45}$$

Letting $\rho = \langle A, \Pi(-)A\rangle_{\mathcal{H}_\circ} : \mathrm{Borel}(\mathbb{R}^2) \to [0,\infty)$ denote the spectral measure associated to A, the right-hand side above is

$$\langle A, TA\rangle_{\mathcal{H}_\circ} = \int_{\mathbb{R}^2} e^{-\omega\delta t + ip\delta x}\,\mathrm{d}\rho(\omega,p). \tag{46}$$

Our goal is to show that ρ is supported in the region $\{\omega \geq |p|\}$. Because the support of the spectral measure does not change upon translating A forward in time, it suffices to replace A by a forward translate by a fixed $b > 0$. Since our tests are compactly supported, by choosing b large enough we may arrange that A is supported in $\{t > |x|\}$.

Then A and $J_\Delta J_\circ A$ are both tests in the second Osterwalder–Schrader Hilbert space $\mathcal{H}_\Delta$. Thus, if T maps the half space $\{t + x > 0\}$ to itself, then

$$\mathbb{E}_\mu[(J_\circ A)TA] = \mathbb{E}_\mu[J_\Delta(J_\Delta J_\circ A)TA] = \langle J_\Delta J_\circ A, TA\rangle_{\mathcal{H}_\Delta} \tag{47}$$

may be interpreted as an inner product in $\mathcal{H}_\Delta$. The condition that $T_{\delta t,\delta x}$ map $\{t + x > 0\}$ to itself says that $\delta t + \delta x > 0$. Let $v = t + x$, $u = t - x$ denote "light wedge" coordinates. In $\mathcal{H}_\Delta$, the Hamiltonian H_Δ is the generator of translations in v, and P_Δ is the generator of translations in u. Hence

$$T = \int_{\mathbb{R}^2} e^{-\omega(\delta t+\delta x)+ip(\delta t-\delta x)}\,\mathrm{d}\Pi_\Delta(\omega,p) = \int_{\mathbb{R}^2} e^{(ip-\omega)\delta t-(ip+\omega)\delta x}\,\mathrm{d}\Pi_\Delta(\omega,p) \tag{48}$$

where $\Pi_\Delta : \mathrm{Borel}(\mathbb{R}^2) \to \mathcal{B}(\mathcal{H}_\Delta)$ is the spectral measure for the pair (H_Δ, P_Δ). So, letting $\rho_\Delta = \langle J_\Delta J_\circ A, \Pi_\Delta A\rangle_{\mathcal{H}_\Delta}$ (this is a complex measure), we have

$$\langle J_\Delta J_\circ A, TA\rangle_{\mathcal{H}_\Delta} = \int_{\mathbb{R}^2} e^{(ip-\omega)\delta t-(ip+\omega)\delta x}\,\mathrm{d}\rho_\Delta(\omega,p). \tag{49}$$

In summary,

$$\int_{\mathbb{R}^2} e^{-\omega\delta t+ip\delta x}\,\mathrm{d}\rho(\omega,p) = \mathbb{E}_\mu[(J_\circ A)TA] = \int_{\mathbb{R}^2} e^{(ip-\omega)\delta t-(ip+\omega)\delta x}\,\mathrm{d}\rho_\Delta(\omega,p), \tag{50}$$

This holds for all $\delta t > 0$ and $\delta x > -\delta t$. For the argument below, it is somewhat simpler to work with

$$\delta v = \delta t + \delta x, \quad \delta u = \delta t - \delta x. \tag{51}$$

Then, the identity above reads

$$\int_{\mathbb{R}^2} e^{\frac{1}{2}(ip-\omega)\delta v-\frac{1}{2}(\omega+ip)\delta u}\,\mathrm{d}\rho(\omega,p) = \int_{\mathbb{R}^2} e^{-\omega\delta v+ip\delta u}\,\mathrm{d}\rho_\Delta(\omega,p). \tag{52}$$

In order to extract information about the measures ρ, ρ_Δ, we integrate both sides of this equality against a function of the form $h(\delta v)\overline{h(\delta u)}$ for (possibly complex-valued) $h \in L^1((0,\infty))$. Recall that

$$\mathcal{L}h(z) = \int_0^\infty e^{-zs}h(s)\,\mathrm{d}s \tag{53}$$

denotes the Laplace transform of h. We now compute

$$\int_0^\infty \int_0^\infty h(\delta v)\overline{h(\delta u)} \int_{\mathbb{R}^2} e^{\frac{1}{2}(ip-\omega)\delta v - \frac{1}{2}(\omega+ip)\delta u}\,\mathrm{d}\rho(\omega,p)\,\mathrm{d}\delta u\,\mathrm{d}\delta v = \int_{\mathbb{R}^2} \Big|\mathcal{L}h\Big(\frac{\omega-ip}{2}\Big)\Big|^2\,\mathrm{d}\rho(\omega,p) \tag{54}$$

$$\int_0^\infty \int_0^\infty h(\delta v)\overline{h(\delta u)} \int_{\mathbb{R}^2} e^{-\omega\delta v + ip\delta u}\,\mathrm{d}\rho_\Delta(\omega,p)\,\mathrm{d}\delta u\,\mathrm{d}\delta v = \int_{\mathbb{R}^2} \mathcal{L}h(\omega)\overline{\mathcal{L}h(ip)}\,\mathrm{d}\rho_\Delta(\omega,p). \tag{55}$$

Thus,

$$\begin{aligned} 0 \le \int_{\mathbb{R}^2} \Big|\mathcal{L}h\Big(\frac{\omega-ip}{2}\Big)\Big|^2\,\mathrm{d}\rho(\omega,p) &= \int_{\mathbb{R}^2} \mathcal{L}h(\omega)\overline{\mathcal{L}h(ip)}\,\mathrm{d}\rho_\Delta(\omega,p) \\ &\le \|\rho_\Delta\| \sup_{\omega\ge 0}|\mathcal{L}h(\omega)| \sup_{p\in\mathbb{R}}|\mathcal{L}h(ip)|. \end{aligned} \tag{56}$$

By Cauchy–Schwarz, $\|\rho_\Delta\| \le \|J_\Delta J_\circ A\|_{\mathcal{H}_\Delta}\|A\|_{\mathcal{H}_\Delta} < \infty$.

Now the game is to find h such that $\mathcal{L}h$ is not too large on $(0,\infty)\cup i\mathbb{R}$ but $\mathcal{L}h$ is large at $(\omega-ip)/2$ whenever $|p|>\omega$. Take $h(s) = s^{iF}e^{-\varepsilon s}$, where $\varepsilon>0$ and $F\in\mathbb{R}$ are parameters. This satisfies

$$\mathcal{L}h(z) = \int_0^\infty s^{iF}e^{-s(z+\varepsilon)}\,\mathrm{d}s = C\Big(1+\frac{z}{\varepsilon}\Big)^{-1-iF} \tag{57}$$

for some numerical constant $C = C(\varepsilon,F)\in\mathbb{C}^\times$. Thus,

$$|\mathcal{L}h(z)| = |C|\cdot\Big|1+\frac{z}{\varepsilon}\Big|^{-1} e^{F\arg(\varepsilon+z)}, \tag{58}$$

Dividing out $|C|^2$, the inequality above reads

$$\int_{\mathbb{R}^2}\Big|1+\frac{\omega-ip}{2\varepsilon}\Big|^{-2} e^{2F\arg(\varepsilon+z)}\,\mathrm{d}\rho(\omega,p) \le \|\rho_\Delta\| e^{\pi|F|/2}. \tag{59}$$

for $z=(\omega-ip)/2$. Take first $F\to+\infty$. On a sector $\arg(\varepsilon+z)\ge\pi/4+\eta$, the left-hand side grows at least as $e^{(\pi/2+2\eta)F}$ times its weighted ρ-mass, whereas the right-hand side is $O(e^{\pi F/2})$. That sector therefore has zero ρ-measure.* Taking $F\to-\infty$ excludes $\arg(\varepsilon+z)<-\pi/4$. Thus $|p|\le\omega+2\varepsilon$ on the support of ρ. Taking $\varepsilon\downarrow 0$, we conclude $\omega\ge|p|$. The spectral support is unchanged by the preliminary forward translation, since its spectral weight is multiplied by the strictly positive function $e^{-2b\omega}$. Polynomial classes with compact positive-time supports are dense, so the forbidden spectral projection vanishes on all of $\mathcal{H}_\circ$. □

5. Extending reflection positivity

Reflection positivity is preserved under certain sorts of holomorphic deformations of the reflection form. This is a one-variable form of Glaser's positive-kernel continuation principle [Gla74, Theorem A]; we give the Taylor–Gram proof in the form needed here:

Lemma 5.1. Let $\mathcal{X}$ denote a complex vector space equipped with a map $j:\mathcal{X}\to L^2_+$, and let $O = O^*\subset\mathbb{C}$ denote a connected open domain in the complex plane containing the origin. Say we have chosen, for each pair of complex numbers $\xi,\zeta\in O$, a sesquilinear form $\langle -,-\rangle_{\zeta,\xi}:\mathcal{X}^2\to\mathbb{C}$ such that

- for each $A,B\in\mathcal{X}$, the function $(\zeta,\xi)\mapsto\langle A,B\rangle_{\zeta,\xi}$ is jointly holomorphic on O^2;
- for each $X\in\mathcal{X}$ there exists a $\delta>0$ and smooth curve $[0,\delta)\ni s\mapsto X(s)\in L^2_+$ such that $X(0)=j(X)$ and
$$\langle A,B\rangle_{r,s} = \langle A(r),B(s)\rangle_\circ \tag{60}$$
for all $r,s>0$ small enough.

*The weight $|1+z/\varepsilon|^{-2}$ is strictly positive. Restrict the nonnegative integral to the sector, divide by $e^{\pi F/2}$, and let $F\to\infty$. A sequence $\eta\downarrow 0$ excludes the whole open sector.

Then, for each $z \in O$, the form $\langle -, - \rangle_{z,\bar{z}}$ is positive semidefinite:

$$\langle A, A \rangle_{z,\bar{z}} \geq 0 \tag{61}$$

for all $A \in \mathcal{X}$. ■

Here $O^* = \{\bar{z} : z \in O\}$, and the forms are antilinear in their first entry, consistently with (41).

Proof idea. Fix $A \in \mathcal{X}$. To warm up, we show that $\langle A, A \rangle_{z,\bar{z}} \geq 0$ for z near the origin. Via analyticity, we may Taylor expand

$$\langle A, A \rangle_{\zeta,\xi} = \sum_{a=0}^{\infty} \sum_{b=0}^{\infty} \frac{\zeta^a \xi^b}{a!b!} \partial_r^a \partial_s^b \langle A, A \rangle_{r,s} \Big|_{r=s=0} \tag{62}$$

for ζ, ξ near 0. The series on the right-hand side is absolutely convergent. Now plug in $\zeta = z$ and $\xi = \bar{z}$:

$$\begin{aligned} \langle A, A \rangle_{z,\bar{z}} &= \sum_{a=0}^{\infty} \sum_{b=0}^{\infty} \frac{z^a \bar{z}^b}{a!b!} \partial_r^a \partial_s^b \langle A, A \rangle_{r,s} \Big|_{r=s=0} \\ &= \lim_{N \to \infty} \sum_{a,b=0}^{N} \frac{z^a \bar{z}^b}{a!b!} \partial_r^a \partial_s^b \langle A, A \rangle_{r,s} \Big|_{r=s=0}. \end{aligned} \tag{63}$$

We can write the quantity under the limit as

$$\sum_{a,b=0}^{N} \frac{z^a \bar{z}^b}{a!b!} \partial_\zeta^a \partial_\xi^b \langle A, A \rangle_{\zeta,\xi} \Big|_{\zeta=\xi=0} = v^\dagger M[N](0) v \tag{64}$$

for $v = (v_0, \ldots, v_N) \in \mathbb{C}^{N+1}$ the vector whose jth component is $v_j = \bar{z}^j / j!$ and $M[N](z)$ the generally complex matrix

$$M[N](z) = \begin{bmatrix} \langle A, A \rangle_{\zeta,\xi} & \cdots & \partial_\xi^N \langle A, A \rangle_{\zeta,\xi} \\ \vdots & \ddots & \vdots \\ \partial_\zeta^N \langle A, A \rangle_{\zeta,\xi} & \cdots & \partial_\zeta^N \partial_\xi^N \langle A, A \rangle_{\zeta,\xi} \end{bmatrix} \Bigg|_{\zeta=z, \xi=\bar{z}}. \tag{65}$$

The (a, b)th entry of $M[N](z)$ is

$$M[N]_{a,b}(z) = \partial_\zeta^a \partial_\xi^b \langle A, A \rangle_{\zeta,\xi} \Big|_{\zeta=z, \xi=\bar{z}}, \qquad 0 \leq a, b \leq N. \tag{66}$$

If we know that $M[N](0)$ is positive semidefinite, then $\langle A, A \rangle_{z,\bar{z}} \geq 0$ follows. To see that $M[N](0)$ is positive semidefinite, write

$$\begin{aligned} M[N]_{a,b}(0) &= \partial_r^a \partial_s^b \langle A, A \rangle_{r,s} \Big|_{r=s=0} \\ &= \partial_r^a \partial_s^b \langle A(r), A(s) \rangle_\circ \Big|_{r=s=0} \\ &= \langle \partial_r^a A(r), \partial_s^b A(s) \rangle_\circ \Big|_{r=s=0} = \langle A^{(a)}(0), A^{(b)}(0) \rangle_\circ. \end{aligned} \tag{67}$$

Thus, for every $v \in \mathbb{C}^{N+1}$,

$$v^\dagger M[N](0) v = \Big\langle \sum_{n=0}^{N} v_n A^{(n)}(0), \sum_{n=0}^{N} v_n A^{(n)}(0) \Big\rangle_\circ \geq 0, \tag{68}$$

since $\langle -, - \rangle_\circ$ is positive semidefinite, as desired.* □

*One-sided derivatives at zero suffice. Smoothness of the curves and continuity of the reflection pairing justify differentiation; the derivatives remain in the closed space L^2_+.

Proof. Fix A, and consider the subset $\Omega \subset O$ consisting of $z \in O$ such that, for each $N \geq 0$, the complex matrix $M[N](z)$ defined above is positive semidefinite. Our goal is to show that $\Omega = O$. Indeed, if $M[N](z)$ is positive semidefinite, then

$$\langle A, A\rangle_{z,\bar{z}} = \mathbf{e}^\dagger M[N](z)\mathbf{e} \geq 0 \tag{69}$$

for $\mathbf{e} = (1, 0, \ldots, 0)$.

Because the matrix $M[N](z)$ depends continuously on z, the set Ω is closed in O. The sketch above contains the full proof that $0 \in \Omega$, so Ω is nonempty. Next, we show that Ω is open, hence all of O since O is connected. This is the general complex analytic fact in Theorem D, applied to the holomorphic function $F(\zeta, \xi) = \langle A, A\rangle_{\zeta,\xi}$. □

Theorem D. Let $O \subset \mathbb{C}$ be open and invariant under complex conjugation, and let $F : O^2 \to \mathbb{C}$ be holomorphic. For $N \geq 0$ define

$$M_F[N](z)_{a,b} = \partial_\zeta^a \partial_\xi^b F(\zeta, \xi)\Big|_{\zeta=z,\xi=\bar{z}}, \qquad 0 \leq a, b \leq N. \tag{70}$$

Then the set

$$\Omega_F = \{z \in O : M_F[N](z) \text{ is positive semidefinite for every } N \geq 0\} \tag{71}$$

is open in O. ■□

Proof. Suppose $z_0 \in \Omega_F$. Taylor expand

$$\partial_\zeta^a \partial_\xi^b F(\zeta, \xi) = \sum_{\alpha,\beta\geq 0} \frac{(\zeta - z_0)^\alpha (\xi - \bar{z}_0)^\beta}{\alpha!\beta!} \partial_\zeta^{a+\alpha} \partial_\xi^{b+\beta} F(z_0, \bar{z}_0) \tag{72}$$

on a sufficiently small polydisc about $(z_0, \bar{z}_0)$. This series is absolutely convergent. If z is sufficiently close to z_0, it follows that

$$M_F[N](z) = \lim_{L\to\infty} Z_L M_F[N + L](z_0) Z_L^\dagger, \tag{73}$$

where Z_L is the $(N + 1)$-by-$(N + L + 1)$ matrix with entries

$$(Z_L)_{a,c} = \begin{cases} (z - z_0)^{c-a}/(c - a)!, & 0 \leq c - a \leq L, \\ 0, & \text{otherwise.} \end{cases} \tag{74}$$

Indeed, the (a, b) entry of the matrix under the limit is

$$\sum_{\alpha,\beta=0}^{L} \frac{(z - z_0)^\alpha (\bar{z} - \bar{z}_0)^\beta}{\alpha!\beta!} \partial_\zeta^{a+\alpha} \partial_\xi^{b+\beta} F(z_0, \bar{z}_0), \tag{75}$$

which converges to $M_F[N](z)_{a,b}$. Each $Z_L M_F[N + L](z_0) Z_L^\dagger$ is the congruence of a positive semidefinite matrix, hence positive semidefinite itself. Therefore, for every $v \in \mathbb{C}^{N+1}$,

$$v^\dagger M_F[N](z) v = \lim_{L\to\infty} (Z_L^\dagger v)^\dagger M_F[N + L](z_0)(Z_L^\dagger v) \geq 0. \tag{76}$$

The same polydisc works for every N, since all derivatives of F are holomorphic there. Thus every $M_F[N](z)$ is positive semidefinite for z in one common neighborhood of z_0. Consequently $z \in \Omega_F$, and Ω_F is open. □

6. The analytic continuation

For $z = (t, x) \in \mathbb{C}^2$ let

$$q(z) = \Re t - |\Im x|, \qquad \mathcal{W} = \{z : q(z) > 0\}. \tag{77}$$

By Proposition 4.1, the translation operators extend to

$$V(z) = \int e^{-t\omega + ixp} \, \mathrm{d}\Pi(\omega, p). \tag{78}$$

They are contractions on $q \geq 0$, strongly continuous there and operator-norm holomorphic on $\mathcal{W}$. Indeed, on $q \geq \delta > 0$ a spectral derivative of total order k is bounded by $\omega^k e^{-\delta\omega} \leq k!\delta^{-k}$, since $|p| \leq \omega$. The semigroup law still holds, and $V(z)\Omega = V(z)^*\Omega = \Omega$.

6.1. **Analyticity across a separating hyperplane.** We prove Lemma 2.1. Translate its separating hyperplane to time zero, using its coordinate normal as the time direction. Choose $\delta > 0$ and the supports sufficiently close to the given configuration that the negative group lies in $t < -\delta$ and the positive group in $t > \delta$. Let A_- and A_+ denote the products of fields on the negative and positive sides. Put

$$\psi_- = [U(-\delta\mathbf{n})JA_-], \qquad \psi_+ = [U(-\delta\mathbf{n})A_+], \tag{79}$$

where J is reflection in this hyperplane. Both are positive-side classes, and reflection and translation give

$$\mathbb{E}_\mu[A_- U(s\mathbf{n} + \mathbf{v})A_+] = \langle \psi_-, V(2\delta + s, \mathbf{v})\psi_+ \rangle. \tag{80}$$

For complex fields the first class is formed using the antilinear J, so the displayed identity is still bilinear in the original field entries. The ordinary pullback in the lemma is $T^*_{-s\mathbf{n}-\mathbf{v}}$, exactly the translation $U(s\mathbf{n} + \mathbf{v})$ here.

For a coordinate normal, repeat the cone argument with each other coordinate as the transverse direction. The joint spectral variables then satisfy $|p_i| \leq \omega$ for every tangential coordinate. Consequently the right side of (80) is holomorphic whenever

$$2\delta + \Re s > \sum_i |\Im v_i|. \tag{81}$$

This proves the claimed real analyticity for $s > -2\delta$ and all real $\mathbf{v}$. Under the four planar hypotheses the same argument uses only the other planar momentum, proving the last assertion of the lemma. If one of the groups is empty the moment is constant by translation invariance. This completes the proof of Lemma 2.1.

6.2. **A field in a time slab.** We now construct moment kernels with their transverse variables smeared. Rescale F by $(2C)^{-1/2}$ so that

$$\mathbb{E}_\mu e^{F(\chi)} \leq e^{\|\chi\|_2^2/2}. \tag{82}$$

This does not change the symmetries or the conclusion. Below K is a numerical constant which may increase between estimates.

The first estimate keeps a source in a time slab before differentiating it. The dyadic reflection iteration is the one used for semigroup bounds by Seiler–Simon [SS76, Theorem 4.3, (23)–(24)]. Their Theorem 7.1, (31)–(33), duplicates reflected sources, and Theorem 8.2 combines repeated reflections with Cauchy estimates. Here the Gaussian bound supplies the quantitative source control. Let h be a complex L^2 test supported in $0 \leq t \leq l$, and let A be a bounded positive-time observable. Then

$$\|[e^{F(h)}U(le_0)A]\| \leq e^{\|\Re h\|_2^2/2}\|[A]\|. \tag{83}$$

To prove this, reflection followed by translation gives

$$\|[e^{F(h)}U(le_0)A]\|^2 = \langle [A], [e^{F(\tau_{le_0}(Jh+h))}U(2le_0)A] \rangle, \tag{84}$$

where on tests $Jh = \Theta_\circ^* \bar{h}$. Define

$$h_0 = h, \quad h_{r+1} = \tau_{2^r l e_0}(Jh_r + h_r), \quad a_r = \|[e^{F(h_r)} U(2^r l e_0) A]\|. \tag{85}$$

The source h_r is supported in $0 \le t \le 2^r l$ and $\|\Re h_r\|_2^2 = 2^r \|\Re h\|_2^2$, since the adjacent slab interiors are disjoint. Cauchy–Schwarz gives

$$a_r^2 \le \|[A]\| a_{r+1}, \qquad a_0 \le \|[A]\|^{1-2^{-r}} a_r^{2^{-r}}. \tag{86}$$

On the other hand, the defining reflection form and (82) give

$$a_r^2 \le \|A\|_\infty^2 \exp(2^r \|\Re h\|_2^2). \tag{87}$$

Substitution and $r \to \infty$ prove (83). If $[A] = 0$, the first Cauchy–Schwarz inequality already gives $a_0 = 0$.

For χ supported in $|t| \le \delta$, define

$$B_\delta(\chi)[A] = [F(\tau_{\delta e_0}\chi) U(2\delta e_0) A]. \tag{88}$$

Apply Cauchy's estimate to the entire Hilbert-valued map $z \mapsto [e^{zF(\tau_{\delta e_0}\chi)} U(2\delta e_0) A]$ on $|z| = \|\chi\|_2^{-1}$. Its power series converges first in ordinary $L^2(\mu)$, by the Gaussian bound. Equation (83) therefore gives

$$\|B_\delta(\chi)\| \le K \|\chi\|_2. \tag{89}$$

The estimate kills null vectors, so this is a bounded operator on $\mathcal{H}_\circ$.* Reflection and translation give

$$B_\delta(\chi)^* = B_\delta(J\chi), \tag{90}$$

$$V(r+t,x) B_\delta(\chi) V(r-t,-x) = B_{\delta+r}(\tau_{(t,x,0)}\chi), \qquad r \ge |t|. \tag{91}$$

For example, both sides of (91) applied to $[A]$ equal $[F(\tau_{(\delta+r+t,x,0)}\chi) U(2(\delta+r)e_0) A]$. If χ is supported in $|t-c| \le \delta_0$ and $|c| + \delta_0 \le \delta$, the same identity gives

$$B_\delta(\chi) = e^{-(\delta+c-\delta_0)H} B_{\delta_0}(\tau_{-ce_0}\chi) e^{-(\delta-c-\delta_0)H}. \tag{92}$$

6.3. **Localizing an insertion in the plane.** Fix a transverse profile $f \in L^2(\mathbb{R}^{d-2}; \mathcal{T}_\mathbb{C})$. The notation includes $d = 2$, when there is no transverse variable. For planar $z = (t,x)$ put

$$j_\varepsilon(z) = (\pi\varepsilon^2)^{-1} 1_{\{|z|<\varepsilon\}}, \qquad A_{f,\varepsilon}(\xi,\eta) = V(\xi - \varepsilon e_0) B_\varepsilon(j_\varepsilon f) V(\eta - \varepsilon e_0). \tag{93}$$

This is defined for $q(\xi), q(\eta) > \varepsilon$. The slab estimate alone loses ε^{-1}. Holomorphy replaces this shrinking scale by the fixed neighboring gap. Write $q_0 = \min(q(\xi), q(\eta))$. We claim

$$\|A_{f,\varepsilon}(\xi,\eta)\| \le \frac{K\|f\|_2}{q_0 - \varepsilon}. \tag{94}$$

Put $r = (q_0 - \varepsilon)/2$ and, for unit vectors u, v, consider $h_\varepsilon(\zeta) = \langle u, A_{f,\varepsilon}(\xi + \zeta, \eta - \zeta) v \rangle$. It is holomorphic for $|\Re\zeta_t| < r$, $|\Im\zeta_x| < r$. Write $\zeta = z + ib$, with $z, b \in \mathbb{R}^2$. For $|b_x| < r$ the outer factors

$$V(\xi - (r+\varepsilon)e_0 + ib), \qquad V(\eta - (r+\varepsilon)e_0 - ib) \tag{95}$$

are contractions, independent of z. Between them, (91) identifies the remaining operator as $B_{r+\varepsilon}((\tau_z j_\varepsilon) f)$ for $z \in (-r,r)^2$. Thus for $\alpha \in L^2((-r,r)^2)$,

$$\left| \int \alpha(z) h_\varepsilon(z + ib) \, \mathrm{d}z \right| \le K\|f\|_2 \|\alpha * j_\varepsilon\|_2 \le K\|f\|_2 \|\alpha\|_2. \tag{96}$$

Duality bounds the real-square integral of $|h_\varepsilon(z+ib)|^2$ by $K^2\|f\|_2^2$. The mean-value inequality on a complex polydisc of radius $r' < r$, followed by integration over the imaginary square, gives

$$|h_\varepsilon(0)|^2 \le (\pi r'^2)^{-2} (2r)^2 K^2 \|f\|_2^2. \tag{97}$$

Let $r' \uparrow r$ and take the supremum over u, v to prove (94).

*Truncating A in $L^4(\mu)$ extends (88) to the products of fields used below. The field L^4 bound and Hölder's inequality justify the passage to the quotient.

We next identify the limit as $\varepsilon \downarrow 0$. Fix $\delta = q_0/2$, take $0 < \varepsilon < \delta/2$, and let α be supported in $(-\delta/2, \delta/2)^2$. Translation covariance gives

$$\int \alpha(z) A_{f,\varepsilon}(\xi + z, \eta - z)\, \mathrm{d}z = V(\xi - \delta e_0) B_\delta((\alpha * j_\varepsilon) f) V(\eta - \delta e_0). \tag{98}$$

The right side converges in operator norm because $\alpha * j_\varepsilon \to \alpha$ in L^2. For fixed vectors u, v, apply Montel's theorem to $\langle u, A_{f,\varepsilon}(\xi + \zeta, \eta - \zeta) v\rangle$. These scalar functions are locally uniformly bounded by (94). The norm limit in (98) fixes the integral of every subsequential limit against every test on the real square. Thus any two limits agree there, and holomorphic uniqueness makes them agree throughout the polydisc. Every sequence $\varepsilon \downarrow 0$ therefore has the same scalar limit at $\zeta = 0$. The limits are bounded sesquilinear forms in u, v, hence define a bounded operator $A_f(\xi, \eta)$ satisfying

$$\|A_f(\xi, \eta)\| \le \frac{K \|f\|_2}{\min(q(\xi), q(\eta))}. \tag{99}$$

It is complex-linear in f. This insertion is localized only in the plane; its transverse variables remain smeared.

The translation and adjoint identities pass to the weak limit:

$$V(\kappa) A_f(\xi, \eta) V(\kappa') = A_f(\kappa + \xi, \eta + \kappa'), \tag{100}$$

$$A_f(\xi, \eta)^* = A_{Jf}(-\Theta_\circ \bar{\eta}, -\Theta_\circ \bar{\xi}). \tag{101}$$

Here $q(\kappa), q(\kappa') \ge 0$, and $Jf = \rho(\Theta_\circ)\bar{f}$ acts only on indices. The shift identity also proves norm holomorphy: near (ξ_0, η_0), choose small $a > 0$ and factor the operator as

$$V(\xi - \xi_0 + a e_0) A_f(\xi_0 - a e_0, \eta_0 - a e_0) V(\eta - \eta_0 + a e_0). \tag{102}$$

For compactly supported $\alpha \in L^2(\mathbb{R}^2)$ with $|t| < \delta$ on its support, we recover the smeared insertion as

$$B_\delta(\alpha f) = \int \alpha(z) A_f(\delta e_0 + z, \delta e_0 - z)\, \mathrm{d}z. \tag{103}$$

The integral converges in operator norm by (99). To verify the identity, take ε smaller than half the distance of the support to $|t| = \delta$. Equation (91) gives the regularized identity

$$\int \alpha(z) A_{f,\varepsilon}(\delta e_0 + z, \delta e_0 - z)\, \mathrm{d}z = B_\delta((\alpha * j_\varepsilon) f). \tag{104}$$

The right side converges in norm to $B_\delta(\alpha f)$. On the left, the matrix elements converge under a uniform bound on the compact support, giving (103). We shall also split inner gaps without changing a product: if $a, b, a', b' \in \mathcal{W}$ and $a + b = a' + b'$, then

$$A_f(\xi, a) A_g(b, \eta) = A_f(\xi, a') A_g(b', \eta). \tag{105}$$

Indeed, subtract a sufficiently small δe_0 from all four inner arguments and use (100). Both sides become $A_f(\xi, \delta e_0) V(a + b - 2\delta e_0) A_g(\delta e_0, \eta)$.

6.4. **The ordered moment kernels.** First separate the test variables:

$$\chi_j(t, x, y) = \alpha_j(t, x) f_j(y), \qquad f_j \in L^2(\mathbb{R}^{d-2}; \mathcal{T}_\mathbb{C}). \tag{106}$$

Finite sums of such products are dense in L^2, also with prescribed planar supports. For $N \ge 2$ and $Z = (z_1, \dots, z_N)$ with $\Delta_j = z_{j+1} - z_j \in \mathcal{W}$, define

$$S_N(Z; f) = \langle \Omega, A_{f_1}(\kappa, \Delta_1/2) \Big[\prod_{j=2}^{N-1} A_{f_j}(\Delta_{j-1}/2, \Delta_j/2) \Big] A_{f_N}(\Delta_{N-1}/2, \kappa) \Omega \rangle. \tag{107}$$

The product is ordered by increasing j and is the identity if $N = 2$. The choice of $\kappa \in \mathcal{W}$ does not matter: the shift identity and $V\Omega = V^*\Omega = \Omega$ let any two choices be enlarged to a common positive-time value. Put $S_0 = 1$ and $S_1 = 0$, consistently with the mean-zero field.

These kernels are holomorphic in the points, invariant under common translations, and complex multilinear in the profiles. Taking κ with a sufficiently large time component in (99) proves

$$|S_N(Z;f)| \le K^N \prod_j \|f_j\|_2 \left(\min_{j<N} q(z_{j+1}-z_j)\right)^{-N}. \tag{108}$$

For compact planar supports strictly ordered in time, they represent the original moments:

$$S_N(\alpha_1 f_1,\dots,\alpha_N f_N) = \int \prod_j \alpha_j(z_j) S_N(Z;f)\,\mathrm{d}Z. \tag{109}$$

To check the identity, enclose the supports in disjoint slabs $(c_j-\delta_j, c_j+\delta_j)$, ordered in j, and translate all of them into positive time. Equations (88) and (92) express the class of the product as

$$e^{-(c_1-\delta_1)H} B_{\delta_1}(\tau_{-c_1 e_0}\chi_1)\Big[\prod_{j=2}^{N} e^{-(c_j-\delta_j-c_{j-1}-\delta_{j-1})H} B_{\delta_j}(\tau_{-c_j e_0}\chi_j)\Big]\Omega. \tag{110}$$

Insert (103). Adjacent semigroups combine into $z_{j+1}-z_j$, and (105) divides each gap equally. Pairing with Ω gives (109). All gaps are positive, so the integrands are uniformly bounded on the compact supports. No product of fields at coincident points is used.

6.5. **Joining the ordered domains.** Permuting the entries gives every time ordering. Quarter turns give the corresponding orderings with either t or x as time. On a quarter-turned domain, rotate the points to the original domain and apply the inverse pullback to their profiles. The kernels agree on overlaps: the intersection is convex, and with every point it contains its real part. A nonempty intersection therefore contains a real open set, where both kernels give the original moments by (109). Testing against products of smooth functions and then applying the holomorphic identity theorem gives equality on the intersection.

We continue to denote the joined kernel by $S_N(Z;f)$. Its real domain contains the configurations for which either all times or all x coordinates are distinct. It is symmetric under simultaneous permutations of points and profiles. With $\varrho(\zeta)=\rho(R(\zeta))$ acting on the tensor indices, quarter-turn symmetry reads

$$S_N(R(-\pi/2)Z;\varrho(\pi/2)f) = S_N(Z;f). \tag{111}$$

This is the discrete symmetry already deduced from reflection positivity, not the continuous covariance being proved.

7. The growth estimate on narrow product tests

The preceding reduction leaves us to prove covariance on narrow product tests. We establish the quantitative estimate needed for that task:

$$|D_N[\chi](\theta+i\beta)| \le C_\chi e^{3N|\beta|}.$$

For these tests the estimate holds for *all* real θ,β. Small real angles suffice in the later propagation of $D_N=0$, but the Fourier contour shift here needs the entire complex cylinder.

We retain the hypotheses of Theorem B. The ordered kernels and the insertion estimates have already been constructed in §6; in particular, (108) gives

$$|S_N(Z;f)| \le K^N \left(\min_{j<N} q(z_{j+1}-z_j)\right)^{-N} \prod_{j=1}^{N} \|f_j\|_2.$$

There is one inverse planar gap per field. The transverse variables remain smeared, so the ambient dimension does not enter this power. We use these ordered domains throughout the proof.

Fix a narrow planar configuration. Choose an ordering, a unit vector n, and $\alpha<\pi/4$ so that all successive displacements lie within angle α of n. Choose small product neighborhoods with compact

closures so that the same ordering, direction n, and angle α work throughout them. Consider real product tests

$$\chi_j(t,x,y) = \alpha_j(t,x) f_j(y)$$

with planar supports in these neighborhoods. Narrowness is an open condition.

Proposition 7.1. If the planar tensor rotation weights have absolute value at most two, then for each such narrow product tuple the discrepancy has an entire, 2π-periodic continuation satisfying

$$|D_N[\chi](\theta + i\beta)| \le C_\chi e^{3N|\beta|} \qquad (\theta, \beta \in \mathbb{R}). \tag{112}$$

Consequently $c_\ell[\chi] = 0$ for $|\ell| > 3N$. ∎

Proof. First cover every real angle. Write $n = R(\phi)(1,0)$. For any real θ, choose an integer k with $|\theta - \phi - k\pi/2| \le \pi/4$. In the corresponding quarter-turn frame, all real gaps point within angle $\alpha + \pi/4 < \pi/2$ of positive time. Thus

$$q(R(-\theta + k\pi/2)(z_{j+1} - z_j)) \ge \cos(\alpha + \pi/4)|z_{j+1} - z_j|. \tag{113}$$

The frame may change with θ, but this bound does not. On the compact product support its right side has a common positive lower bound, denoted by δ.

Then extend each chart through all imaginary heights. The identity

$$q(z) = \min\{\Re(z_t + iz_x), \Re(z_t - iz_x)\}$$

and the rotation convention give

$$(R(-i\beta)z)_t + i(R(-i\beta)z)_x = e^{-\beta}(z_t + iz_x), \qquad (R(-i\beta)z)_t - i(R(-i\beta)z)_x = e^{\beta}(z_t - iz_x).$$

Consequently

$$q(R(-i\beta)z) \ge e^{-|\beta|} q(z) \qquad (q(z) > 0). \tag{114}$$

In particular, imaginary rotations preserve the ordered domain.

For the kth frame the strict real-angle condition is

$$|\Re\zeta - \phi - k\pi/2| < \pi/2 - \alpha. \tag{115}$$

On this whole vertical strip, define the continued discrepancy by

$$\int \prod_j \alpha_j(z_j) S_N\big(R(-\zeta + k\pi/2)Z; \varrho(\zeta - k\pi/2) f\big)\, dZ - S_N(\chi). \tag{116}$$

Here ϱ acts on the profiles' tensor indices only, as in (111). The scalar tests are always evaluated at real points. Every gap in this integral lies in the ordered domain, by (114). On compact subsets of the strip the gaps have a positive uniform margin, so the kernel bounds justify the integral and its holomorphy.

For real ζ, (116) equals the actual discrepancy: apply the known quarter-turn covariance to the moment, and then use the ordered smearing identity and a change of variables. This uses no covariance under an arbitrary real rotation. The strips (115) cover $\mathbb{C}$ and overlap, since their width $\pi - 2\alpha$ exceeds their center spacing $\pi/2$. Any overlap contains a real interval, where the two formulas equal the same actual discrepancy. The one-variable identity theorem makes the formulas agree on the entire overlap. They therefore glue to an entire function. Replacing ζ by $\zeta + 2\pi$ and k by $k + 4$ leaves (116) unchanged, proving periodicity.

Estimate the glued function. For $\zeta = \theta + i\beta$, choose the nearest frame used in (113). Combining the two gap estimates gives

$$q(R(-\theta - i\beta + k\pi/2)(z_{j+1} - z_j)) \ge \delta e^{-|\beta|}.$$

The N inverse gaps in equation (108) therefore contribute at most $\delta^{-N} e^{N|\beta|}$.

This is where representation theory enters. In an invariant norm, the complexified planar rotation splits into weight spaces with factors $e^{im\theta}$. If $|m| \le 2$, then

$$\|\varrho(\theta + i\beta - k\pi/2)f_j\|_2 \le e^{2|\beta|}\|f_j\|_2. \tag{117}$$

The real angle and the real quarter turn add no norm cost. The N profiles contribute at most $e^{2N|\beta|}$, independently of the chosen frame. Equation (108) now gives

$$\left|S_N(R(-\theta - i\beta + k\pi/2)Z; \varrho(\theta + i\beta - k\pi/2)f)\right| \le K^N\delta^{-N}\prod_j \|f_j\|_2\, e^{3N|\beta|}. \tag{118}$$

Integrate against the fixed scalar tests in (116). For example, a permissible constant in (112) is

$$C_\chi = K^N\delta^{-N}\prod_j \|\alpha_j\|_1\|f_j\|_2 + |S_N(\chi)|.$$

It is independent of θ, β. Finite linear combinations of narrow product expressions of the same order are handled by adding these constants.

Finally, use the *entire cylinder* just constructed. With

$$c_\ell[\chi] = \frac{1}{2\pi}\int_0^{2\pi} D_N[\chi](\theta)e^{-i\ell\theta}\,d\theta,$$

Cauchy's theorem on the rectangle between heights 0 and β gives

$$\begin{aligned} c_\ell[\chi] &= \frac{1}{2\pi}\int_0^{2\pi} D_N[\chi](\theta + i\beta)e^{-i\ell\theta}e^{\ell\beta}\,d\theta, \\ |c_\ell[\chi]| &\le C_\chi e^{3N|\beta|+\ell\beta}. \end{aligned} \tag{119}$$

Periodicity cancels the vertical sides. Send $\beta \to +\infty$ when $\ell < -3N$ and $\beta \to -\infty$ when $\ell > 3N$. This proves the cutoff. For $N = 0, 1$ the conclusion is immediate from $S_0 = 1$ and $S_1 = 0$; the latter follows from Gaussian domination. □

Quarter-turn symmetry makes the real discrepancy $\pi/2$-periodic; holomorphic uniqueness gives the same periodicity for its entire continuation. Hence only $\ell \in 4\mathbb{Z}$ can occur. Since $D_N[\chi](0) = 0$, the cutoff gives, on these narrow tests,

$$D_N[\chi](\zeta) = \sum_{\substack{\ell\in 4\mathbb{Z} \\ 0<|\ell|\le 3N}} c_\ell[\chi](e^{i\ell\zeta} - 1). \tag{120}$$

These are the original Fourier coefficients from (37). The moment bound makes each coefficient a bounded multilinear form on L^2 tests, so product density extends (120) to all tests with the same small compact planar supports. The later positivity comparisons eliminate the remaining frequencies on narrow configurations. The separated-group analyticity reduction then propagates $D_N[\chi](\theta) = 0$.

7.1. **Testing on narrow configurations.** To apply reflection positivity while staying on narrow configurations, write

$$\mathcal{C}_+ = \{(t, x) \in \mathbb{R}^2 : t > |x|\}. \tag{121}$$

We use real product tests $\chi_j = \alpha_j f_j$, with $\alpha_j \in C_c^\infty(\mathbb{R}^2; \mathbb{R})$ and real transverse L^2 profiles. Call an ordered tuple *admissible* if its compact planar supports $K_j = \operatorname{supp}\alpha_j$ satisfy

$$K_j \subset \mathcal{C}_+, \qquad K_{j+1} - K_j \subset \mathcal{C}_+. \tag{122}$$

The empty tuple is included. All inequalities are strict and hold for every choice of points in the supports.

For two admissible tuples χ, η with planar supports K_j, L_k, the reflected concatenation $(J\chi, \eta)$ is narrow. Reverse the first block to order it from negative to positive time. Its internal gaps are the

original gaps acted on by $-\Theta_\circ$, which preserves $\mathcal{C}_+$. When both tuples are nonempty, the crossing gap satisfies

$$z_1' - \Theta_\circ z_1 = z_1' + (-\Theta_\circ z_1) \in \mathcal{C}_+, \qquad z_1 \in K_1, \quad z_1' \in L_1, \tag{123}$$

since both summands lie in $\mathcal{C}_+$. Thus every successive gap lies in $\mathcal{C}_+$, including for squared norms and mixed pairings of different admissible tuples. A finite subdivision of the compact supports therefore makes (120) available for all these pairings.

Lemma 7.2. If $\ell \neq 0$ and $c_\ell^{(N)}$ is nonzero near a narrow configuration, then for every $a + b = N$ it is detected on a tuple $(J\chi, \eta)$ with χ, η admissible of lengths a, b. ■

Proof. The defect identity

$$D_N[R(\phi)^*\chi](\theta) = D_N[\chi](\theta + \phi) - D_N[\chi](\phi) \tag{124}$$

gives, directly from the Fourier integral,

$$c_\ell[R(\phi)^*\chi] = e^{i\ell\phi} c_\ell[\chi], \qquad \ell \neq 0. \tag{125}$$

Common translations and permutations also preserve the coefficients. Localize a nonzero coefficient at a narrow configuration, rotate its common narrow direction to the time direction, and order its planar points $z_1, \dots, z_N$ so that $z_{j+1} - z_j \in \mathcal{C}_+$. These operations preserve nonvanishing. If both groups are nonempty, choose the origin at the midpoint of z_a, z_{a+1}. For $j \leq a$ the vector from z_j to the midpoint is a half-gap plus a sum of $\mathcal{C}_+$ gaps; for $j > a$ the vector from the midpoint to z_j has the same property. Consequently the first group lies in $-\mathcal{C}_+$ and the second in $\mathcal{C}_+$. Reflect and reverse the first group. Both tuples now satisfy (122) on sufficiently small product neighborhoods, where detecting real product tests can be chosen.* If one group is empty, put the origin sufficiently far forward or backward in time instead. This proves the claim. □

7.2. **The continued reflection pairing.** We now apply Lemma 5.1 to the actual moment kernels. Let $A = \prod_{j=1}^a F(\chi_j)$ and $B = \prod_{k=1}^b F(\eta_k)$ be monomials with admissible tuples. In particular, their supports are strictly ordered in $t > 0$; the two positive tuples may overlap each other. For real r, s small enough, put

$$A(r) = \prod_j F(R(r)^*\chi_j), \qquad B(s) = \prod_k F(R(s)^*\eta_k). \tag{126}$$

These are smooth L^2_+ curves by the moment estimates. We take formal finite linear combinations of such monomials and extend the forms sesquilinearly, conjugating the first coefficient. It suffices to work with any fixed finite family.†

Write $\chi_j = \alpha_j f_j$ and $\eta_k = \gamma_k g_k$. Define $\langle A, B\rangle_{\zeta,\xi}$ by integrating the ordered kernel with its first group of points and profiles

$$R(\zeta)\Theta_\circ Z, \quad \varrho(-\zeta)Jf, \qquad \text{and its second group} \qquad R(-\xi)Z', \quad \varrho(\xi)g, \tag{127}$$

against $\prod_j \alpha_j \prod_k \gamma_k$. List the reflected first group in reverse order. Here $Jf = \rho(\Theta_\circ)\bar{f}$, and ϱ is precisely the tensor pullback in (111). The relation $\varrho(-r)\rho(\Theta_\circ) = \rho(\Theta_\circ)\varrho(r)$ fixes the signs in the first group. For small real r, s the density identity gives

$$\langle A, B\rangle_{r,s} = \langle A(r), B(s)\rangle_\circ. \tag{128}$$

*At a point in the support of the coefficient distribution, the Schwartz kernel theorem detects a nonzero value on products in arbitrarily small product neighborhoods. Approximation by finite sums of $\alpha(t,x)f(y)$ in L^2, followed by multilinearity and the moment bound, gives the required product tests. Real and imaginary parts let them be chosen real.

†Formal monomials give a vector space with the natural map j to L^2_+; no choice of representation of an L^2 observable is needed. All positivity and Cauchy–Schwarz assertions use only finite families.

For the fixed finite family choose $\delta > 0$ so that real rotations of width δ preserve every within-group time gap and the gap from each positive support to the origin. The kernel integral is jointly holomorphic for

$$(\zeta, \xi) \in O^2, \qquad O = \{z : |\Re z| < \delta\}. \tag{129}$$

Indeed, the reflected points lie in $-\mathcal{W}$ and the positive points in $\mathcal{W}$. Independent imaginary rotations preserve these cones and the within-group gaps; the crossing gap is a sum of two elements of $\mathcal{W}$. On compact subsets of O^2 the insertion bounds are uniform on the supports, justifying integration. Thus every hypothesis of Lemma 5.1 is satisfied.

At $(\zeta, \xi) = (-i\beta, i\beta)$ the two groups in (127) are the same imaginary rotation of the original reflected configuration. Their profiles are $\varrho(i\beta)Jf$ and $\varrho(i\beta)g$. For real angles near zero the integral agrees with the rotated moment. Holomorphic uniqueness in the vertical strip, together with (120), gives

$$\langle A, B\rangle_\beta := \langle A, B\rangle_{-i\beta, i\beta} = S_{a+b}(J\chi, \eta) + D_{a+b}[J\chi, \eta](i\beta), \qquad \beta \in \mathbb{R}. \tag{130}$$

Here $J\chi$ is the reflected, conjugated tuple. Its concatenation with η is narrow by (123), which is why the finite continuation (120) applies. The form on the left is positive semidefinite. Empty tuples give the vacuum and are included in this statement. Positivity applies to the continued moment in (130).

8. Eliminating Fourier coefficients

The use of parity in eliminating extreme transformation degrees follows Streater's method, as recorded by Osipov [Str72; Osi77]. We give the comparison in the present Fourier variable.

We now finish the proof of Theorem C. Let M_N be the largest $|\ell|$ for which c_ℓ is nonzero near some narrow configuration, and set it to zero if all nonzero frequencies vanish there. Then

$$M_0 = 0, \qquad M_N \in 4\mathbb{Z}, \qquad 0 \le M_N \le 3N. \tag{131}$$

The conjugation identity exchanges positive and negative frequencies, so whenever $M_N > 0$ the coefficient at $-M_N$ is nonzero as well. These coefficients are invariant under common translations and permutations of the entries. Translation invariance follows from $R(\theta)^*\tau_a = \tau_{R(-\theta)a}R(\theta)^*$ in both terms of the defect.

8.1. $2M_{a+b} \le M_{2a} + M_{2b}$.

Fix nonnegative integers a, b. If $M_{a+b} = 0$, there is nothing to prove, so assume $M_{a+b} > 0$. By Lemma 7.2, its negative coefficient is detected on a tuple $(J\chi, \eta)$ with χ, η admissible of lengths a, b, so that

$$c_{-M_{a+b}}[J\chi, \eta] \neq 0. \tag{132}$$

Equation (130) and the narrow finite expansion give

$$\langle A, B\rangle_\beta = \big(c_{-M_{a+b}}[J\chi, \eta] + o(1)\big)e^{M_{a+b}\beta} \qquad (\beta \to +\infty). \tag{133}$$

Similarly, $\langle A, A\rangle_\beta = O(e^{M_{2a}\beta})$ and $\langle B, B\rangle_\beta = O(e^{M_{2b}\beta})$. All three tuples are narrow by the admissibility conditions. The form is positive semidefinite, so Cauchy–Schwarz gives

$$|\langle A, B\rangle_\beta|^2 \le \langle A, A\rangle_\beta \langle B, B\rangle_\beta. \tag{134}$$

Since the leading coefficient in (133) is nonzero, comparison of the exponents yields

$$2M_{a+b} \le M_{2a} + M_{2b}. \tag{135}$$

8.2. $M_{2a} \in 8\mathbb{Z}$**.** Suppose that $M = M_{2a} > 0$. Multiplying the positive form $\langle -, - \rangle_\beta$ on admissible degree-a monomials by $e^{-M\beta}$ and letting $\beta \to +\infty$ gives a positive semidefinite form

$$(A, B) \longmapsto c_{-M}[J\chi, \eta]. \tag{136}$$

The testing lemma supplies a nonzero mixed value. Applying Cauchy–Schwarz to this limiting form shows that some diagonal value is strictly positive:

$$c_{-M}[J\chi, \chi] > 0. \tag{137}$$

This is a sign assertion about a leading squared-norm coefficient; mixed Fourier coefficients need not have a sign.

The remainder of the argument studies the effect of a 45° rotation $Q = R(\pi/4)$ on the measure. Define $\tilde{\mu}$ by

$$\tilde{\Phi}(\chi) = \Phi(Q^*\chi), \qquad \tilde{S}_N(\chi) = S_N(Q^*\chi). \tag{138}$$

The four planar mirrors are permuted by this change of frame. Reflection invariance and positivity, the Gaussian estimate and translation invariance are preserved. These are all the hypotheses used in the chosen plane, so the same argument applies to $\tilde{\mu}$, in its own Hilbert space.

Since planar rotations commute,

$$\tilde{D}_N[\chi](\theta) = D_N[\chi](\theta + \pi/4) - D_N[\chi](\pi/4). \tag{139}$$

Matching the nonzero Fourier coefficients gives

$$\tilde{c}_\ell[\chi] = e^{i\ell\pi/4} c_\ell[\chi] = (-1)^{\ell/4} c_\ell[\chi], \qquad \ell \in 4\mathbb{Z} \setminus \{0\}. \tag{140}$$

Nonvanishing is unchanged, so $\tilde{M}_N = M_N$. Apply positivity in the two frames to the same tuple in (137):

$$0 \le \tilde{c}_{-M}[J\chi, \chi] = (-1)^{M/4} \underbrace{c_{-M}[J\chi, \chi]}_{>0}. \tag{141}$$

Thus $(-1)^{M/4} = 1$, which means $M \in 8\mathbb{Z}$. The zero case is automatic, so

$$M_{2a} \in 8\mathbb{Z} \qquad (a \ge 0). \tag{142}$$

This completes the proof, because:

Lemma 8.1. Suppose $0 = M_0, M_1, M_2, \dots$ are nonnegative integers satisfying

$$M_n \le 3n, \qquad 2M_{a+b} \le M_{2a} + M_{2b}, \qquad M_{2n} \in 8\mathbb{Z} \tag{143}$$

for all nonnegative integers a, b, n. Then $M_n = 0$ for every n. ■

Proof. Note that $0 \le M_2 \le 6$ and $M_2 \in 8\mathbb{Z}$, so $M_2 = 0$. Define $\delta_n = M_{2n} - M_{2n-2}$. Applying convexity with $a = n - 1$ and $b = n + 1$ yields

$$2M_{2n} \le M_{2n-2} + M_{2n+2}, \tag{144}$$

hence $\delta_n \le \delta_{n+1}$. The increments are nondecreasing multiples of eight and $\delta_1 = 0$, so all are nonnegative. If δ_r were positive, it would be at least eight, as would every later increment. This implies

$$M_{2j} \ge M_{2r-2} + 8(j - r + 1) \ge 8(j - r + 1), \tag{145}$$

contradicting $M_{2j} \le 6j$ for large j. Therefore every increment vanishes, and $M_{2n} = 0$. Finally, $0 \le 2M_n \le M_0 + M_{2n} = 0$ controls the remaining terms. □

Remark. Convexity and the divisibility $M_{2n} \in 8\mathbb{Z}$ constrain the even-order increments: once one is positive, every later increment is at least eight. This is incompatible with the growth bound $M_N \le 3N$.

9. Conclusion

The preceding sections prove $M_N = 0$ for every N, so (120) gives $D_N[\chi](\theta) = 0$ near every narrow configuration. This proves Theorem C. The narrowing and small-angle arguments give Theorem B; the transverse-cut reduction then gives Theorem A for rotations of the tx-plane. The same proof applies to every coordinate plane. These rotations generate $\mathrm{SO}(d)$, and the coordinate reflections already preserve the law, so the conclusion holds for all of $\mathrm{O}(d)$.

For compactly supported tests with pairwise disjoint supports this is precisely the distributional vanishing off Δ established above. General Schwartz tuples with pairwise disjoint supports follow by compact truncation, which preserves disjointness, and (14). Restoring the original normalization of F does not change the covariance identity. The theorem concerns these off-diagonal moments; no removal of contact terms has been used.

10. Sharpness of the conclusion and the weight restriction

The hypotheses permit noncovariant contact terms. Moreover, the representation restriction cannot be enlarged uniformly to include planar weight three. Both limitations already occur in dimension two.*

10.1. **Noncovariant contact terms.** Let $\mathcal{T} = \mathrm{Sym}^2_0(\mathbb{R}^2)$, with orthonormal basis

$$e_1 = \frac{dt^2 - dx^2}{\sqrt{2}}, \qquad e_2 = \frac{dt \otimes dx + dx \otimes dt}{\sqrt{2}}.$$

Take independent scalar white noises in these two components, with variances $a, b > 0$, where $a \neq b$. Their characteristic function is

$$\Phi(\chi) = e^{-Q(\chi)/2}, \qquad Q(\chi) = \int_{\mathbb{R}^2} (a\chi_1^2 + b\chi_2^2).$$

The law is translation invariant and satisfies $\mathbb{E}e^{F(\chi)} \leq e^{\max(a,b)\|\chi\|_2^2/2}$.

In this basis the coordinate reflections act by $\mathrm{diag}(1,-1)$ and the diagonal reflections by $\mathrm{diag}(-1,1)$. They preserve Q. If the tests lie on one side of any required mirror, their reflected supports are disjoint from the original supports. Hence

$$\Phi(\chi_j - \Theta^*\chi_k) = e^{-Q(\chi_j)/2} e^{-Q(\chi_k)/2},$$

a rank-one positive semidefinite matrix. Thus all the required reflection positivities hold.

The tensor pullback of $R(\theta)$ is $R(-2\theta)$. At $\theta = \pi/4$ it exchanges e_1, e_2, up to sign. Taking $\chi = fe_1$ shows that $Q(\chi) = a\|f\|_2^2$, whereas $Q(R(\pi/4)^*\chi) = b\|f\|_2^2$. The law is therefore not rotation invariant. Its covariance kernel is $\mathrm{diag}(a,b)\delta(q-q')$. Every nonempty moment on pairwise disjoint supports nevertheless vanishes, since the centered Gaussian evaluations are independent. The failure of covariance is entirely supported at coincidences.

10.2. **Failure at planar weight three.** Let ϕ be the massive scalar Gaussian free field on $\mathbb{R}^2$, with covariance $(-\Delta + m^2)^{-1}$, $m > 0$. Its covariance is reflection positive, and the associated Gaussian characteristic function is reflection positive; see [Uhl79, pp. 119–120].

Let $\mathcal{T}_3$ be the two-dimensional space of real harmonic cubics. Choose an orthonormal basis (e_1, e_2) proportional to

$$t^3 - 3tx^2, \qquad 3t^2x - x^3.$$

For the clockwise rotations used here,

$$\rho_3(R(\theta)) = R(-3\theta), \qquad \rho_3(\Theta_\circ) = \mathrm{diag}(-1,1).$$

Thus the planar weights are ± 3. Put $S = \mathrm{diag}(1,-1)$ and define the $\mathcal{T}_3$-valued field by

$$F = (\partial_t\phi, -\partial_x\phi), \qquad F(\chi) = \phi(L\chi), \qquad L\chi = -\operatorname{div}(S\chi).$$

*The two examples in this section were supplied in the referee report.

For every symmetry g of the square,

$$S\rho_3(g) = g^{\mathsf{T}} S.$$

Indeed, this holds for a quarter turn and for $\Theta_\circ$, which generate the group; the pullback multiplication order preserves the identity. Since $gS\rho_3(g) = S$, the chain rule gives

$$L(g^*\chi) = (L\chi) \circ g.$$

The map L preserves supports and commutes with translations. Consequently scalar reflection positivity transfers to the characteristic function of F across all four required mirrors. The law is translation invariant.

With unitary Fourier transform,

$$\mathbb{E}F(\chi)^2 = \int_{\mathbb{R}^2} \frac{|p_t\widehat{\chi}_1 - p_x\widehat{\chi}_2|^2}{|p|^2 + m^2}\,\mathrm{d}p \le \|\chi\|_2^2.$$

Thus $\mathbb{E}e^{F(\chi)} \le e^{\|\chi\|_2^2/2}$ for every real test, with the required uniform L^2 bound.

It remains to check that covariance fails away from coincidences. Write $h(r)$ for the scalar covariance at positive separation. The matrix-valued covariance kernel is

$$K(z) = -S(D^2h)(z)S.$$

For two first-component insertions initially separated by $(r, 0)$, the pullback rotates the separation to $rn_\theta = R(-\theta)(r, 0)$ and the profile to $v_\theta = \rho_3(R(\theta))e_1$. In particular,

$$n_\theta = (\cos\theta, \sin\theta), \qquad Sv_\theta = (\cos 3\theta, -\sin 3\theta), \qquad (Sv_\theta)\cdot n_\theta = \cos 4\theta.$$

The radial Hessian formula therefore yields

$$v_\theta^{\mathsf{T}} K(rn_\theta)v_\theta = -\frac{1}{2}\left(h''(r) + \frac{h'(r)}{r}\right) - \frac{1}{2}\left(h''(r) - \frac{h'(r)}{r}\right)\cos 8\theta.$$

Its difference from the value at $\theta = 0$ is

$$\frac{1}{2}\left(h''(r) - \frac{h'(r)}{r}\right)(1 - \cos 8\theta).$$

The coefficient is strictly positive. Indeed,

$$h(r) = \int_0^\infty \frac{e^{-m^2s - r^2/(4s)}}{4\pi s}\,\mathrm{d}s, \qquad h''(r) - \frac{h'(r)}{r} = \int_0^\infty \frac{r^2}{4s^2}\frac{e^{-m^2s - r^2/(4s)}}{4\pi s}\,\mathrm{d}s > 0.$$

For $r > 0$, exponential decay at both ends justifies differentiating under the integral. At $\theta = \pi/8$ the defect is nonzero. The kernel is smooth away from the origin, so sufficiently small disjoint smooth tests detect this defect in the actual two-point moment.

This example has $M_2 = 8$. Allowing weight three would replace the angular bound by $M_N \le 4N$, which permits this value. Hence the cutoff two is sharp for a theorem covering every representation in the stated class: the next integer cutoff already fails in dimension two.

11. A scalar lattice application

Consider a subsequential continuum probability limit ν of the normalized scalar lattice fields in Sokal [Sok82, §2.3, Lemma 2.2 and Corollary 2.3]. The uniform Gaussian L^2 hypothesis follows from his estimates by the following calculation; it is not a separate lemma stated there.*

*This deduction was supplied in the referee report. We start with the normalized sequences of Sokal's §2.3. His Note Added in Proof, p. 387, says that the proof of Proposition 2.1 is insufficient and refers to [BFS83, Proposition 5.1] for its repair. That construction step concerns the choice of parameters giving the prescribed correlation length. The calculation here verifies the inputs for a given normalized sequence and its probability limits; it does not supply a new proof of the earlier existence or nontriviality construction.

Use the single zero-boundary infinite-volume state of Sokal's §2.1, with its coordinate and diagonal reflection positivities. For lattice spacing a, write

$$F_a(f) = a^d \sum_{i\in\mathbb{Z}^d} f(ai)\varphi_i^{(a)}, \qquad G_a(k) = \mathbb{E}[\varphi_0^{(a)}\varphi_k^{(a)}] \geq 0, \qquad \mathsf{X}_a = \sum_{k\in\mathbb{Z}^d} G_a(k).$$

The susceptibility normalization is

$$\mathsf{X}_a = \widetilde{z}\,\widetilde{m}^{-2}a^{-d}, \qquad \widetilde{z} > 0, \quad \widetilde{m} > 0,$$

with $\widetilde{z}, \widetilde{m}$ fixed. These are the embedding and normalization in [Sok82, (2.81), (2.83)]; $\widetilde{m}$ is the chosen normalization parameter. For real $f \in C_c^\infty(\mathbb{R}^d)$, the exponential estimate [Sok82, (2.27)] gives

$$\mathbb{E}e^{F_a(f)} \leq \exp\left[\frac{a^{2d}}{2}\sum_{i,j}|f(ai)|G_a(i-j)|f(aj)|\right].$$

Translation invariance and $2uv \leq u^2 + v^2$ imply

$$\sum_{i,j}|f(ai)|G_a(i-j)|f(aj)| \leq \mathsf{X}_a \sum_i |f(ai)|^2.$$

Consequently

$$\mathbb{E}e^{F_a(f)} \leq \exp\left[\frac{\widetilde{z}}{2\widetilde{m}^2}a^d\sum_i |f(ai)|^2\right].$$

The exponential-functional convergence in [Sok82, (2.86)] and the Riemann-sum limit yield

$$\mathbb{E}_\nu e^{F(f)} \leq \exp\left[\frac{\widetilde{z}}{2\widetilde{m}^2}\|f\|_2^2\right]. \tag{146}$$

Smooth compact truncation and Fatou's lemma extend this estimate to every real Schwartz test. Thus one constant $C = \widetilde{z}/(2\widetilde{m}^2)$ controls all the tests required by Theorem A.

All the required reflection positivities hold for the same limit. Sokal [Sok82, p. 328, item (l), and p. 329] establishes coordinate site, coordinate bond, and diagonal site reflection positivity simultaneously in the same infinite-volume state. Each coordinate or pair-diagonal reflection through the origin preserves $a\mathbb{Z}^d$, and the scalar embedding commutes with its pullback. A test supported on one open side is sampled only on that side. For every finite test family, (2.86) therefore gives entrywise convergence of its reflection-positive characteristic-functional matrix along the chosen subsequence. The limiting matrix is positive semidefinite. Lemma 2.2 of [Sok82] gives invariance of this same ν under all real translations.

The scalar representation has weight zero, so Theorem A applies: these limits have orthogonally covariant moments away from coincidences. This supplies the rotation implication proposed in Sokal's §5(2). It does not determine arbitrary contact terms. The constant in (146) uses fixed positive $\widetilde{m}$; the calculation makes no assertion about massless limits with divergent susceptibility.

Appendix A. Well-definedness of S_N

Moment estimates from generating-functional analyticity are classical; see Fröhlich [Frö74, Lemma 3.2, pp. 276–277]. For the present Gaussian bound, a tail estimate gives the sharper order dependence used in (14).

For real χ, apply (4) to $u\chi$, $u \in \mathbb{R}$. The exponential Markov inequality, optimized over positive and negative u, gives

$$\mu\{|F(\chi)| > r\} \leq 2\exp\left(-\frac{r^2}{4C\|\chi\|_2^2}\right). \tag{147}$$

Integrating the tail yields

$$\|F(\chi)\|_{L^p(\mu)} \leq K\sqrt{p}\,\|\chi\|_2, \qquad 1 \leq p < \infty, \tag{148}$$

with K depending only on C. The exponential bound near $u = 0$ also gives $\mathbb{E}_\mu F(\chi) = 0$ and $\mathbb{E}_\mu |F(\chi)|^2 \le 2C\|\chi\|_2^2$. Splitting into real and imaginary parts gives (148) for complex tests. Consequently $\chi \mapsto F(\chi)$ extends continuously from $\mathcal{S}$ to L^2 tests, with values in every finite $L^p(\mu)$; the extensions agree. Approximation and Fatou's lemma extend (4) to real L^2 tests.

Hölder's inequality proves (14), hence joint continuity on Schwartz tests. The Schwartz kernel theorem makes S_N a tempered distribution on $(\mathbb{R}^d)^N$, with one tensor index per variable. The characteristic function is given by the absolutely convergent series

$$\Phi(\chi) = \sum_{N=0}^{\infty} \frac{i^N}{N!} S_N(\chi, \dots, \chi). \tag{149}$$

Indeed, $e^{|F(\chi)|} \le e^{F(\chi)} + e^{-F(\chi)}$ is integrable, so dominated convergence applies to the exponential series.

Invariance of the law under an isometry T is therefore equivalent to invariance of every moment on all tests, not merely disjoint ones. Equality of characteristic functions gives equality of the finite-dimensional evaluation laws, and evaluations generate the Borel sigma-algebra of the weak dual.* This establishes the analytic assertions about moments used in the introduction.

Appendix B. RP + translation covariance ⇒ reflection covariance

Fix a hyperplane $H \subset \mathbb{R}^d$.

Proposition B.1. Suppose that $\mu : \mathrm{Borel}(\mathcal{S}') \to [0,1]$ is a Borel probability measure whose law is translation-invariant and reflection positive across H. Then the law is invariant under the reflection Θ across H. That is,

$$\Phi(\chi) = \Phi(\Theta^* \chi). \tag{150}$$

■

Proof. Without loss of generality, it suffices to consider $H = \{t = 0\}$. Because Φ is continuous by dominated convergence, and because compactly supported test functions are dense in $\mathcal{S}$, it suffices to consider real compactly supported χ.

First suppose that χ is supported in $\{t > 0\}$. Reflection positivity applied to $\chi, 0$ says that the matrix

$$M = \begin{pmatrix} \Phi(\chi - \Theta^*\chi) & \Phi(\chi) \\ \Phi(-\Theta^*\chi) & 1 \end{pmatrix} \tag{151}$$

is positive semidefinite: $z^\dagger M z \ge 0$ for all $z \in \mathbb{C}^2$. In particular, $z^\dagger M z \in \mathbb{R}$. This can only hold for all z if M is Hermitian, hence

$$\begin{aligned} \Phi(\chi) = \overline{\Phi(-\Theta^*\chi)} &= \int_{\mathcal{S}'} e^{-iF(-\Theta^*\chi)} \, \mathrm{d}\mu(F) \\ &= \int_{\mathcal{S}'} e^{iF(\Theta^*\chi)} \, \mathrm{d}\mu(F) = \Phi(\Theta^*\chi), \end{aligned}$$

as desired. Here $F(\chi) \in \mathbb{R}$ because these tests and distributions are real.

To extend the conclusion to all compactly supported χ, use translation covariance. Let $T_b : (t, x, y) \mapsto (t + b, x, y)$. If $b > 0$ is large enough that $T_{-b}^* \chi$ is supported in $\{t > 0\}$, then

$$\Phi(\chi) = \Phi(T_{-b}^* \chi) = \Phi(\Theta^* T_{-b}^* \chi) = \Phi(T_b^* \Theta^* \chi) = \Phi(\Theta^* \chi). \tag{152}$$

The first and last equalities use translation invariance of the law. □

*Choose increasing seminorms p_j defining the Schwartz topology and a countable dense set of tests. Each equicontinuous set $E_{j,k} = \{F : |F(\chi)| \le k p_j(\chi) \text{ for all } \chi\}$ is evaluation-measurable, since its inequalities can be checked on that dense set. On $E_{j,k}$ the weak topology is induced by these countably many evaluations and is second countable. Its Borel sets therefore belong to the restricted evaluation sigma-algebra. The countable union of the $E_{j,k}$ is $\mathcal{S}'$, proving the assertion.

AI STATEMENT

This manuscript is a hybrid of AI and human writing.

REFERENCES

[BEG67] J. Bros, H. Epstein, and V. Glaser. "On the connection between analyticity and Lorentz covariance of Wightman functions". *Communications in Mathematical Physics* 6 (1967), 77–100. DOI: 10.1007/BF01654126 (cit. on p. 3).

[BFS83] D. C. Brydges, J. Fröhlich, and A. D. Sokal. "A new proof of the existence and nontriviality of the continuum φ^4_2 and φ^4_3 quantum field theories". *Communications in Mathematical Physics* 91 (1983), 141–186. DOI: 10.1007/BF01211157 (cit. on p. 24).

[Frö74] J. Fröhlich. "Schwinger functions and their generating functionals. I". *Helvetica Physica Acta* 47 (1974), 265–306. DOI: 10.5169/seals-114572 (cit. on pp. 4, 25).

[Gla74] V. Glaser. "On the equivalence of the Euclidean and Wightman formulation of field theory". *Communications in Mathematical Physics* 37 (1974), 257–272. DOI: 10.1007/BF01645941 (cit. on pp. 4, 11).

[JJM13] A. Jaffe, C. D. Jäkel, and R. E. Martinez II. *Complex classical fields and partial Wick rotations*. 2013. arXiv: 1302.5935 [math-ph] (cit. on p. 4).

[MS76] A. Mardin and R. F. Streater. "On the spinor rank of Fermi fields". *Communications in Mathematical Physics* 48 (1976), 81–83. DOI: 10.1007/BF01609412 (cit. on p. 4).

[OS73] K. Osterwalder and R. Schrader. "Axioms for Euclidean Green's functions". *Communications in Mathematical Physics* 31 (1973), 83–112. DOI: 10.1007/BF01645738 (cit. on pp. 1, 2, 4, 9).

[OS75] ——. "Axioms for Euclidean Green's functions II". *Communications in Mathematical Physics* 42 (1975), 281–305. DOI: 10.1007/BF01608978 (cit. on pp. 2, 4, 9).

[Osi77] E. P. Osipov. "Connection between the spectrum condition and the Lorentz invariance of the Yukawa$_2$ quantum field theory". *Communications in Mathematical Physics* 57 (1977), 111–116. DOI: 10.1007/BF01625770 (cit. on pp. 4, 21).

[Sok82] A. D. Sokal. "An alternate constructive approach to the φ^4_3 quantum field theory, and a possible destructive approach to φ^4_4". *Annales de l'Institut Henri Poincaré. Section A* 37 (1982), 317–398. URL: https://www.numdam.org/item/AIHPA_1982__37_4_317_0/ (cit. on pp. 4, 24, 25).

[SS76] E. Seiler and B. Simon. "Nelson's symmetry and all that in the Yukawa$_2$ and $(\phi^4)_3$ field theories". *Annals of Physics* 97 (1976), 470–518. DOI: 10.1016/0003-4916(76)90044-0 (cit. on pp. 4, 14).

[Str72] R. F. Streater. "Connection between the spectrum condition and the Lorentz invariance of $P(\phi)_2$". *Communications in Mathematical Physics* 26 (1972), 109–120. DOI: 10.1007/BF01645698 (cit. on pp. 3, 21).

[Uhl79] A. Uhlmann. "Some remarks on reflection positivity". *Czechoslovak Journal of Physics B* 29 (1979), 117–126. URL: https://www.physik.uni-leipzig.de/~uhlmann/PDF/Uh79c.pdf (cit. on p. 23).